Department of Computer Science

University Of Gujrat

Educational VR Game for Kids



Session: MSc.CS Fall 2019-2021

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Submitted By

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STATEMENT OF SUBMISSION

It is verified that this project titled "**Educational VR game for Kids**" by **Faryal Azhar** D/O **Azhar Iqbal** Roll # (19020819-049), student of Master of Science (Computer science) at Department of Computer Science, University of Gujrat, Pakistan, contains sufficient material required for the award of above said degree.

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Abstract

Since the first time VR technology was used in back 1960s and coined much later in 1987, “Virtual Reality” (VR) has been evolved in many different ways. The VR has acquired great importance in the field of Education especially in adult training and experiments but unfortunately there are not many games/apps available for kids in VR. The main purpose of my project is to develop a interactive game for kids (age group 5-8) which will help them to learn better through experience and in a fun way.

There are two types of VR : Immersive and non-immersive. The latter one allows player to see contents based on device in use, the user is not able to immerse in his environment completely. While on the other hand, Immersive VR allows user to immerse in environment completely which makes them feel that they are inside the environment. I’ll be developing an immersive game by using head-mounted display (HMD) technology (Oculus Quest). The player will be able to experience and indulge into game completely using vr headset and will be able to interact with the objects in game by using controllers which will track the movement of player hands. Moreover, I’ll be developing different levels where player will be able to learn alphabets, counting, colors, mathematical operations and many more. I’ll make the user experience soothing by implementing soft visuals and sound mechanism. With my 1-year experience of game development I’ll make sure that the game runs smooth on device. I’ll be optimizing various settings to improve the performance of game and meanwhile also keep the visuals in context.

Oculus Quest released on 2019 by Oculus (brand by Facebook) is a VR headset which is a standalone device that can run apps and games wirelessly. It have android-based operating system. For developing my game I’ll be integrating oculus package in unity game engine. Unity is an cross-platform game engine used to develop 2d-3d games, animations, films, architecture etc. I’ll be using C-sharp (C#).The game might work on all version and models of Oculus with little or no changes at all, but I’ll be focusing on oculus quest 1 specifically.

VR is based on three basic principles : Immersion, Interaction and user involvement. My game will also follow these principles and will offer very high potential in education by making learning more inspiring, engaging, interactive and fun. It will provide more memorable educational experience and will encourage creative thinking.

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Chapter 1: Project Feasibility Report

1.1. Introduction

Virtual Reality provides the user immersive experience and interaction with the virtually created world. VR is being used in many fields for example: military, Sport, Medical training , Education etc. Virtual Reality (VR) have a great potential in future and also in current world especially since the pandemic has hit the world, creating the disruption in the field of education, business , industrialization and social life of people. The schools, colleges and universities are struggling to provide the education and on the other side parents are worried for their children's education. This is where VR comes in handy.
My main purpose of developing and choosing this particular topic and project was to come up with the solution through which the children of age-group: 5-8 could get the essential education regarding different topics (I.e mathematical operations, spellings etc). I have limited the scope of project to some of the activities due to the shortage of time but it can be expanded further by adding more and more exercises.
My game can be used as a tool for providing the basic education to kids and making the learning experience enjoyable with the factor of immersion and interaction. While playing the game, the user will be asked to solve different educational activities in a fun way. The game will also have various difficulties level so that the activities can be made harder as per user requirement. I made this project by combining the game and education which will encourage the creative thinking in kids.
Game can be played on Oculus quest with hand-controllers (might work on all Oculus VR headset, haven't tested). Moreover, I have used version control git to keep track of my project and to maintain the record of project.

1.2. Project Feasibility Report

There are many types of feasibilities:

- Technical
- Operational
- Economic
- Schedule
- Specification
- Information
- Motivational
- Legal and Ethical

1.2.1. Technical Feasibility

I have take following questions into consideration in technical feasibility:

Q-1: Is the proposed system practical?

Ans: Yes, it is definitely practical. There are a lot of great games in VR but none good educational can be found for kids. In short, the implementation of game is practical as there is a lot of documentation available on internet and all the resources are available.
Q-2: Do I currently possess the necessary technology?

Ans: Yes, I have all the required technology and resources available.
✓ VR headset (Oculus Quest)

- ✓ Unity Game Engine
- ✓ Laptop/PC with required resources
- ✓ C# language
- ✓ Visual Studio
- ✓ Sourcetree (for Git)

1.2.2. Operational Feasibility

Q-1: Whether the problem is worth solving or will the solution to the problem works?
Ans: Yes, off course the problem is worth solving. The VR has great potential in every field of life , whether it is training, social, gaming or education etc. In current world there is urgent need of getting everything accessible online easily and Education is one of the main priority. My solution of developing a game for kids in VR will most probably be very beneficial for kids. They will be able to learn through fun activities most importantly by getting complete immersion and interaction with virtual world.

Q-2: How the end user feel about problem?
Ans: The game is very user friendly. It have easy to use controls and options. Anyone can play this game easily but as it's targeted audience is mainly children. Children can easily play this game whether in the absence or presence of their guardian. Overall, it will provide fun experience to its targeted audience as it is developed with the same intention.

1.2.3. Economical Feasibility

Performing Cost Benefit Analysis (CBA).
Cost for the proposed system (figures in USD).

Table 1.1: Cost Benefit Analysis

Cost/ Year	1	2	3	4	5
Hardware	**1500**				
Software	**500**				
Personnel	**10**	**10**	**15**	**20**	**20**
Maintenance	**0**	**50**	**50**	**50**	**100**
Cost at year end	**2010**	**60**	**65**	**70**	**120**
Cumulative costs	**2010**	**2070**	**2135**	**2205**	**2325**

Benefit for the propose system

Table 1.2: Benefit/year

Benefits / Year	1	2	3	4	5
From finished reports	100	100	100	100	100
Increase in sales	1000	1500	1500	2000	2000
Benefits at year end	1500	1600	1600	2100	2100
Cumulative benefits	1500	3100	4700	6800	8900

Profit = Benefits - Costs
= 8900 - 2325
= 6575 $

Since we are gaining, this project is economical feasible.

1.2.4. Schedule Feasibility

As I'll be developing the project alone, I have scheduled all the task related to my project. According to my plan, the project will be completed within the given time. The meeting with supervisor will also be conducted after implementation of features to get the feedback and improve the feature if required. To pen up the answer, Yes project can be developed within the given time with proper time management.

1.2.5. Specification Feasibility

Requirements :

- ✓ To make the objects in environment interactable and apply proper physics on them.
- ✓ To provide soothing and immersive user experience.
- ✓ To make player learn things by creating different fun activities and levels
- ✓ To implement soft sound/audio mechanism.
- ✓ To minimize motion sickness
- ✓ Make child-friendly UI and controls.
- ✓ To implement animations where ever needed.
- ✓ To design fun theme for game.
- ✓ Optimization of game for better performance.
- ✓ Implementing saving system.

1.2.6. Information Feasibility

The project will be completed within the provided time and going to be very reliable as the importance of VR in STEM is increasing with passage of time. The technology used in project is in trend in present and will be trending even more according to many authentic sources. The targeted audience of this game can get several benefits from it. The project will encourage the creative thinking through many educational fun activities.

1.2.7. Motivational Feasibility

The motivation behind developing this project is to provide the basic education to kids in a completely immersive and interactive environment through fun activities. So while developing the project the view of child will be kept in mind that how they will try to interact with game and what features they will like. Also, the game will be tested by letting a child to play it so that implementation and overall experience could be improved with proper steps.

1.2.8. Legal and Ethical Feasibility

No, the project will not have any legality issues as all the proper rights will be given to everyone. The project contains 3d assets from various artist and sources, which are free for educational purposes. As for now the game will not be published on online store, so the assets are free for educational purposes until and unless not used for any commercial use.
The project will not have any ethical issues as well. The project will not be targeted to any specific ethnicity , it will be available for everyone without any discrimination over race, color, origin, religion etc. Nor it will contain any material which might go against the guideline or hurt anyone's sentiment. Regardless of their religion, ethnicity, color or race everyone will enjoy playing the game. As the game is for kids, it will be the first priority to make it child friendly and avoid using any insensitive material.

1.3. Project Scope

This project was initiated as a mean to provide some sort of educational knowledge to children of age-group 5-7 in a fun way. It will be developed mainly using Unity Game Engine and Oculus. My main focus will be on coding, level designing and testing. The 3d assets will be downloaded from other sources as there is not enough time to design assets. Game will have different fun activities for example: activity related to mathematical operations.

There is a useful acronym– **SMART**– that can help to understand the scope of project more clearly:

Specific:

(Who) The project is meant to be for kids. (especially kids belonging to age group : 5-7)

(What) The project was initiated as a mean to provide some sort of educational knowledge to children of age-group 5-7 in a fun way

(Where) The project (game) can be played anywhere anytime using the Oculus VR headset and controllers.

(When) The project will be accomplished withing the time span of 2-3 months.

(Why) The main focus of project is on concept "Play to Learn". The main idea of the project revolves around the concept of immersion and interaction. Using Virtual Reality, the project will be providing the interactive experience for its player. By playing the game, user will be able to learn things through immersion and interaction and that is the most beneficial thing in any learning process. Moreover, the project will also encourage the creative and problem solving skill in children.

Measurable:

The project's progress will be tracked using different techniques such as GANTT chart.

Achievable:

All the resources needed for the project is available which most importantly includes Oculus headset and controllers. The project will be developed using Unity Game Engine , which is free. All the 3d assets,audio for the game can be downloaded. Visual Studio will be used as code editor, which is also available. Other than that, logo of game will be designed using photoshop which is also available for free usage.
Moreover, with my 1 year of game development experience and strong coding skills, I am totally capable of developing this project withing the given time span.

Realistic:

The project is completely realistic and reachable withing given period of time and resources. I'm totally commit to achieve the said goals and objectives.

Timely:

As the project have deadline withing 3-4 months max, the progress of the project will be tracked properly using different tools which will be very helpful to achieve the goals within the time. Anyhow, project will be surely completed within the given time with proper time management.

Project boundaries:

The Project will only support single player, there will be no 2-player or multiplayer support. As for now, project will not be released on any Online store, as it is a lengthy process and may require more time and quality assurance. The 3d assets, audio will not be designed by project creator as they are very time consuming and out scope for developer. The project will be created within the boundaries of coding, level designing, testing and logo designing. All the other mechanism like animation, sound effects, music , particle effects etc are included in level designing which will be implemented by developer.

Goals and objectives:

- To make the objects in environment interactable and apply proper physics on them. (For example user can hold and interact different objects and all the physics rule will be applicable on object. For instance: if user throw object, object will have gravity).
- To provide soothing and immersive user experience. (Soothing because it is a game for kids. And immersive so that player can totally indulge into game).
- To implement mechanism of joints to doors and drawers. (So the doors and drawers can operate the way they operate in real world).
- To make player learn things by creating different fun activities and levels. (Learn alphabets, colors, spellings, mathematical operation and tables etc).
- To implement soft sound/audio mechanism. (Sound shouldn't be so loud).
- To minimize motion sickness. (motion sickness is one of the major drawback of vr.
- To make the game comfortable for kids, have to minimize motion sickness).
- Make child-friendly UI and controls.
- To implement animations where ever needed.
- To get good graphics by customizing Post-Processing in unity.
- To design fun theme for game.
- Optimization of game for better performance.

1.4. Project Costing

1.4.1. Project Cost Estimation by Function Point Analysis

Number of user I/P = 2
Number of user O/P = 3
Number of user inquiries = 3
Number of user files (logical files) = 1
Number of external interface = 1

Table 1.3: Function Point Analysis

Type of Component	Complexity of Component			
Function Type	**Simple**	**Average**	**Complex**	**Total**
Internal Logical File	1x7=7	1x10=10	1x15=15	32
External Interface File	1x5=5	1x7=7	1x10=10	22
External Input	2x3=6	2x4=8	2x6=12	26
External Output	3x4=12	3x5=15	3x7=21	48
External Inquiry	3x3=9	3x4=12	3x6=18	39

Total number of unadjusted Function Point (UFP)	**167**
Total Adjusted Function Points (FP)	**178.69**

Complexity Adjusted Factor = 3
14-questions
(∑Fi) = 42

FP est. = Count Total x [0.65 + 0.01 x (∑Fi)]
= 167 x [0.65 + 0.01 x 42]
= 167 x [0.65 + 0.42]
= 167 x [1.07]
= 178.69

Cost / FP = labor rate / productivity parameter
= 8000/420
= 12.90 LOC (Line of Code)

Total Project Cost = FP est. * (cost / FP)
= 178.69 * 12.90
= 2305.10

Total Estimated Effort = FP est. / productivity parameter
= 178.69/420
= 0.42

1.4.2. Project Cost Estimation by using COCOMO'81 (Constructive Cost Model)

Basic COCOMO

KLOC= 4

Table 1.4: Basic COCOMO

Type	Effort	Schedule
Organic	PM = 2.4 x (KLOC) $^{1.05}$ = 2.4 x (4)$^{1.05}$ = 2.4 x 4.287 = 10.289	TD = 2.5 x (PM)$^{0.38}$ = 2.5 x (10.289)$^{0.38}$ = 2.5 x 2.42 = 6.05

Intermediate COCOMO

Table 1.5: Intermediate COCOMO

Type	Effort
Organic	PM = 2.4 x (KLOC) $^{1.05}$ x M = 2.4 x (4)$^{1.05}$ x 1.10 = 2.4 x 4.287 x 1.10 = 11.317

The schedule is determined using the Basic COCOMO schedule equations.

Effort = 10.289
Duration = 6.05 months

People Required = Effort / Duration
= 10.289 / 6.05
= 1.70

1.4.3. Activity Based Costing

Table 1.6: Activity Based Costing

Name Activity	Activity	Resources	Duration
A	Planning	Game Design Document	2 weeks
B	Assets collection	3D assets	2 weeks
C	VR connectivity	Oculus package, Unity	1 week
D	UI Designing	Unity	2 week
E	Game/Level Designing	Unity	3 weeks
F	Coding	Unity ,Visual Studio	5 weeks
G	Testing and Debuging	Unity, Oculus	2 week
H	Git management	Source Tree	1 week
I	Logo Designing	Photoshop	1 week

1.5. Task Dependency Table

Table 1.7: Task Dependency Table

Name	Task	Dependency	Duration
A	Planning	None	2 weeks
B	Assets collection	A	2 weeks
C	VR connectivity	A	1 week
D	UI Designing	A,B	2 week
E	Game/Level Designing	A,B,C,D	3 weeks
F	Coding	A,B,C,D,E	5 weeks
G	Testing and Debuging	A,B,C,D,E,F	2 week
H	Git Management	A,B,C,D,E,F,G	1 week
I	Logo Designing	A	1 week

1.6. CPM - Critical Path Method

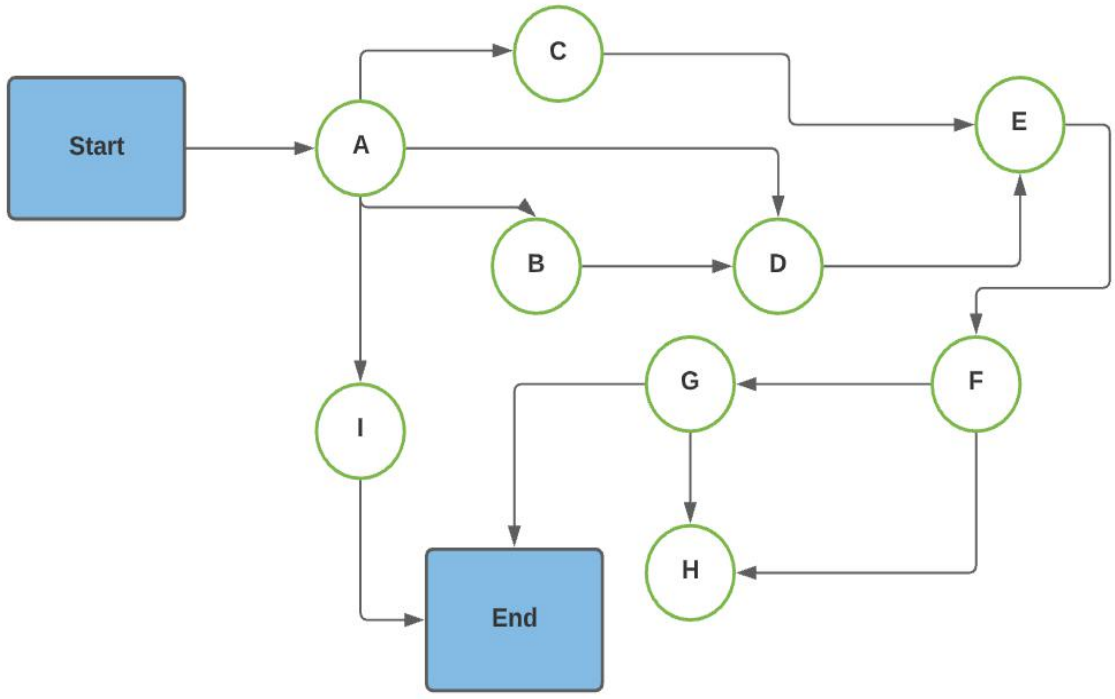


Figure 1.1: Critical Path Method

Network Diagram for the above mentioned activities

Identifying Critical Paths

Table 1.8: Identifying Critical Path

Activity	Duration	ES	EF	LS	LF	TS	FS
A	2	0	2	1	3	1	1
B	2	2	4	3	5	1	2
C	1	2	3	5	6	3	1
D	2	4	6	4	6	0	0
E	3	6	9	6	9	0	0
F	5	9	14	9	14	0	0
G	2	14	16	14	16	0	0
H	1	16	17	16	17	0	0
I	1	2	3	2	3	0	0

The parameters and slacks are calculated as follows:

The critical path is:
D, E, F, G, H, I

1.7. Gantt chart

[Educational VR Game for Kids] Gantt Chart

[Final Year Project]

Project Start Date 1/29/2021 (Friday)

Project Lead Faryal Azhar

WBS	TASK	START	END	WORK DAYS
1	**Idea Hunting**			
1.1	Idea Hunting	Fri 1/29/21	Wed 2/10/21	13
1.2	Meeting with supervisor	Mon 2/15/21	Mon 2/15/21	1
1.3	Discussion with supervisor	Fri 2/19/21	Fri 2/19/21	1
1.4	Checking for resources	Sat 2/20/21	Sun 2/21/21	2
1.5	Finalizing idea	Mon 2/22/21	Mon 2/22/21	1
2	**Proposal Documentation**			
2.1	Project feasibility	Thu 2/25/21	Fri 2/26/21	2
2.2	Project discussion	Mon 3/01/21	Mon 3/01/21	1
2.3	Requirement Analysis	Wed 3/03/21	Fri 3/05/21	2
3	**Session on Project**			
3.1	Meeting with supervisor	Fri 3/05/21	Fri 3/05/21	1
3.2	proposal writing	Mon 3/08/21	Mon 3/15/21	7
3.3	Meeting with supervisor	Thu 3/11/21	Thu 3/11/21	1
3.4	Update Proposal	Fri 3/12/21	Mon 3/15/21	4
3.5	Proposal submission	Tue 3/16/21	Tue 3/16/21	1
4	**Requirement Gathering**			
4.1	Reading resources	Wed 3/17/21	Thu 3/25/21	9
4.2	Meeting with supervisor	Mon 3/22/21	Mon 3/22/21	1
4.3	Complete Requirement	Tue 3/23/21	Mon 3/29/21	6
5	**Design Phase**			
5.1	Use case	Fri 3/26/21	Tue 3/30/21	5
5.2	Data flow diagram	Thu 4/01/21	Sat 4/03/21	3
5.3	Entity relation diagram	Mon 4/05/21	Wed 4/07/21	3
5.4	Finding errors	Thu 4/08/21	Fri 4/09/21	2
5.5	Meeting with supervisor	Mon 4/12/21	Mon 4/12/21	1
5.6	finalizing design	Tue 4/13/21	Sat 4/17/21	5
6	**Implementation**			
6.1	Developement	Sat 4/17/21	Thu 7/01/21	54
6.2	Coding	Mon 4/19/21	Fri 6/25/21	50
6.3	Front end Design	Mon 4/26/21	Wed 5/12/21	13
6.4	Debuging code	Sat 6/19/21	Tue 6/29/21	10
6.5	Refactoring	Sun 6/27/21	Thu 7/01/21	4
7	**Testing**			
7.1	Design Test case	Thu 7/01/21	Fri 7/02/21	2
7.2	Black Box testing	Sat 7/03/21	Sun 7/04/21	1
7.3	White Box testing	Mon 7/05/21	Tue 7/06/21	2
7.4	Error Detection	Wed 7/07/21	Sat 7/10/21	3
7.5	Deployement	Sun 7/11/21	Sun 7/11/21	1
7.6	User Testing	Mon 7/12/21	Wed 7/14/21	3

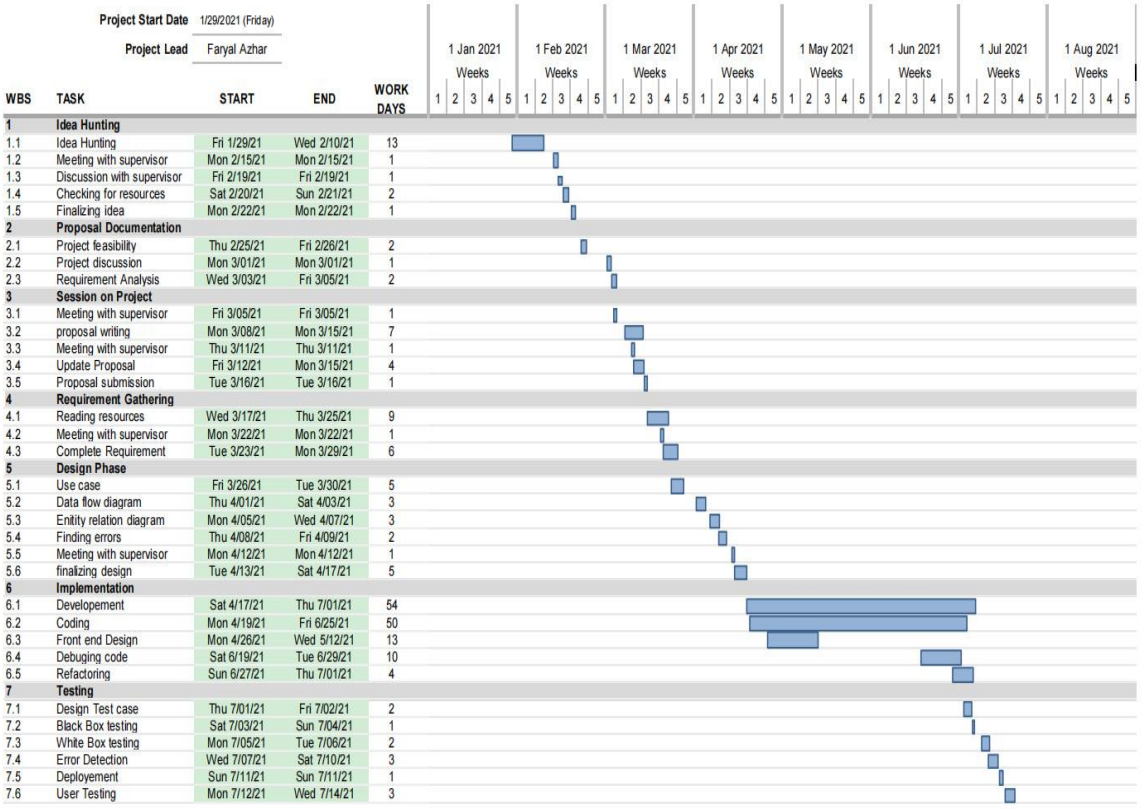


Figure 1.2: Gantt chart

1.8. Introduction to Team member and their skill set

Table 1.9: Introduction to Team member and their skill set

Member Name	Skill Set
Faryal Azhar	➢ Front End Development (Game/Level Designing) ➢ Backend Development (Coding) ➢ Graphic Designing (Logo) ➢ Git Management ➢ Testing / Debuging ➢ Quality Assurance ➢ Code Optimization

1.9. Task and Member Assignment Table

Table 1.10: Task number and name

Task	Task Name
T1	Asset Collection
T2	VR Connectivity
T3	UI designing
T4	Level/Game Designing
T5	Coding
T6	Testing/Debuging
T7	Refactoring
T8	Git Management
T9	Logo Designing

Table 1.11: Task, duration and member assignment

Task	Duration (Days)	Dependencies	Member
T1	12	none	M1
T2	2	T1	M1
T3	15	T1	M1
T4	20	T1, T2	M1
T5	30	T1, T2, T3, T4	M1
T6	7	T5	M1
T7	5	T5	M1
T8	10	T3,T4, T5, T6, T7	M1
T9	4	none	M1

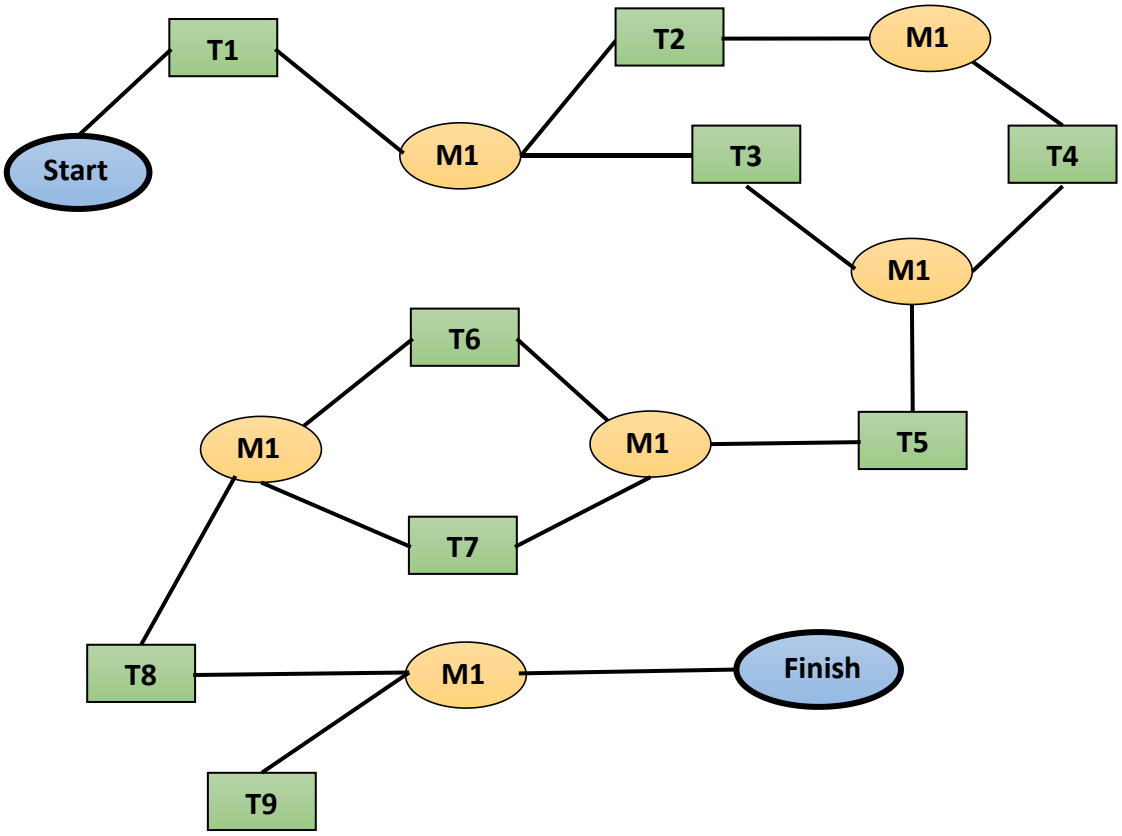


Figure 1.3: Task and Member assignment

1.10. Tools and Technology with reasoning

1.10.1. Front-end Framework

To design the front-end of the game Unity3D Engine will be used. The 3d Assets (3d models) and other UI elements will be either downloaded or designed using Blender if needed. Then they'll be imported in Unity Game engine, after which the developer will scale and arrange them according to the need. The interactive UI will be designed and linked together using Scripts (written in C#).

1.10.2. Back-end Programming Language

The Back-end logic of the game will be written in C-Sharp (C#) language using Visual Studio. The written scripts then will be attached to game objects in unity.

1.10.3. Git Management / Version Control

The Version Control will be very essential part of the project as there are plenty of ways in which things can go wrong. Git will help the developer to restore to the previous working stage, in the case if anything goes wrong .The Git will be managed through "Sourcetree" software. Git will be stored locally and will also be uploaded on Github.

1.10.4. Logo Designing

Being a game, the logo is also very important part of the project.The logo will be designed using “Photoshop”.

1.10.5. VR headset’s Software (Oculus) and Testing

As the game is being developed for Oculus Quest, the Oculus Software will be used to connect the headset with PC. This software is very important, as all the testing will be performed on oculus quest and without this software, the quest cannot be connected with PC.

1.11. Vision Document

The vision behind developing this project is to create a medium using which children can get basic education in a playful and fun way. The VR technology is getting prominent day by day, and it have unlimited usage in many fields. So by using VR technology for this project, it will provide us with the benefit of immersion and interaction. As the studies have shown, the children can learn better with interaction. So, the project will prove very beneficial for everyone (kids, teachers, parents etc).

Checkpoints:

- Have you fully explored what the "problem behind the problem" is?

Ans: Yes, the problem have been explored.

- Is the problem statement correctly formulated?

Ans: Yes

- Is the list of stakeholders complete and correct?

Ans: Yes

- Does everyone agree on the meaning of the system limits?

Ans: Yes

- If system boundaries have been expressed using actors, have all actors been defined and correctly described?

Ans: Yes

- Have you sufficiently explored constraints to be put on the system?

Ans: Yes

- Have you covered all kinds of constraints - for example political, economic, and environmental?

Ans: Yes

- Have all key features of the system been identified and defined?

Ans: Yes

- Will the features solve the problems that are identified?

Ans: Yes

- Are the features consistent with constraints that are identified?

Ans: Yes

1.12. Risks

- **Delay in project's development**
  Any delay in critical path might result in overall project's delay.

- **VR might cause motion-sickness**
  Bad implementation of locomotion might result in motion-sickness. Which can then make the entire game experience unpleasant.

- **Developer's lack of experience**
  Developer's lack of experience in Virtual Reality technology might cause them to stuck into any problem.

- **Unable to create fun and interactive environment in game**
  As the game is for children, making game fun to play is very crucial. If the user don't find overall game experience fun, interactive and is unable to immerse into the environment then the project will be of no use. Also, the theme of game should be fun any soothing.

- **Unable to find appropriate 3d assets/models**
  The use of appropriate 3d assets for designing the environment and UI is very important. There's always a risk that resources to build the project might not be available.

1.13. Product Features / Product Decomposition

- Music and sound mechanism.
- Save and Load Player's progress.
- Sensory Feedback.
- Virtual World with interactivity and immersion.
- Fun activities which will help player learn.
- Teleport and Continuous locomotion for player movement.
- Good graphics.
- Easy to use controls.
- Self-made logo for game.
- Difficulty selection for activities.

Chapter 2: System Requirement Specification

2.1. Introduction:

Requirements engineering process provides the appropriate mechanism for understanding what the customer wants, analyzing need, assessing feasibility, negotiating a reasonable solution, specifying the solution unambiguously, validating the specification and managing the requirements as they are transformed into an operational system. The task of capturing, structuring, and accurately representing the user's requirements so that they can be correctly embodied in systems which meet those requirements (i.e. are of good quality).

- Requirements elicitation
- Requirements analysis and negotiation
- Requirements specification
- System modeling
- Requirements validation
- Requirements management

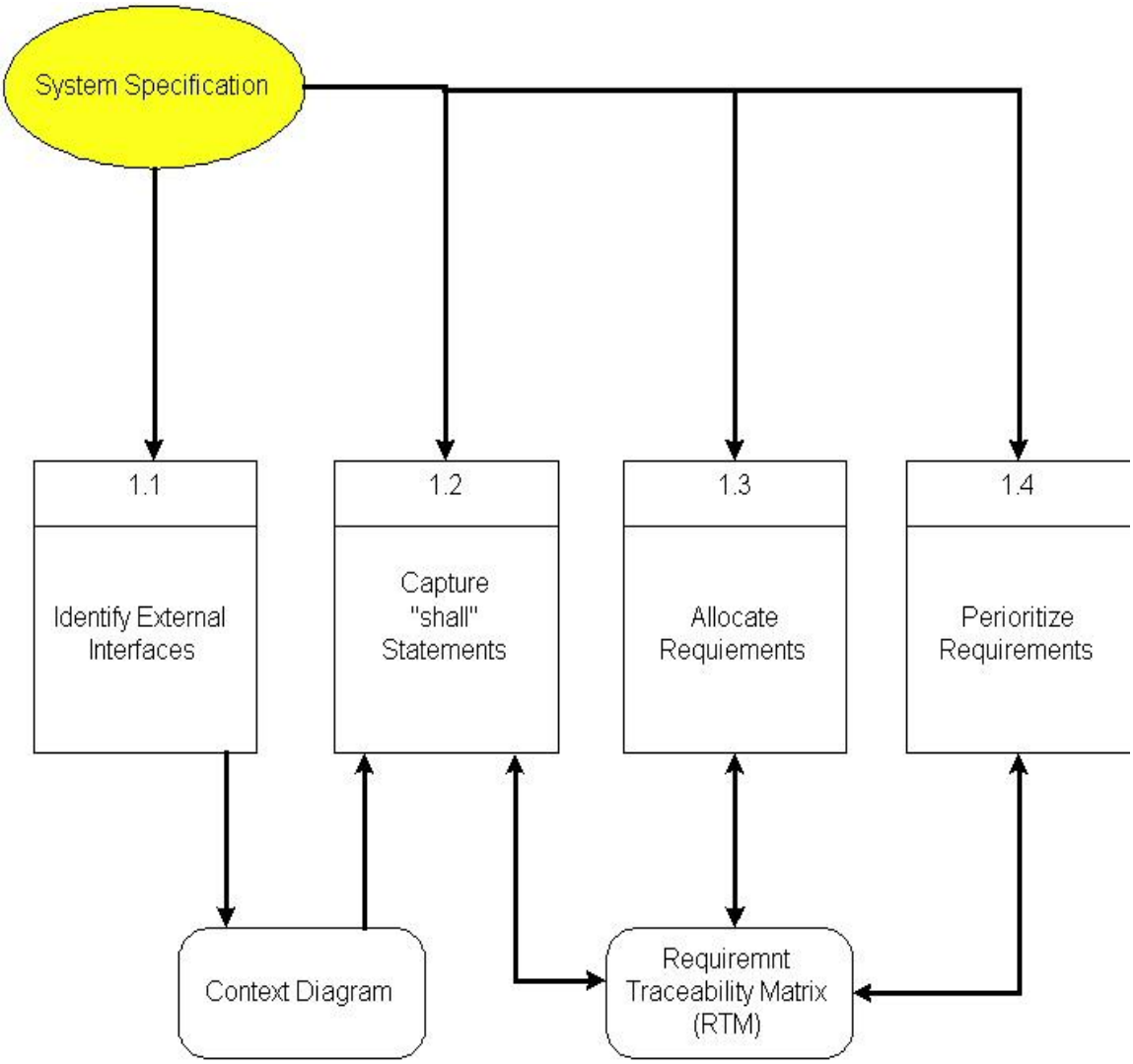


Figure 2.1: Requirement Engineering Process

Here, requirements specification is to be discussed. Requirements specification would lead to the following four steps:

- Identify external interfaces
- Development of context diagram
- Capture “shall statements
- Allocate requirements
- Prioritize requirements
- Development of requirements traceability matrix

2.2. Systems Specification

The following are the clauses that must be included while describing the system specifications.

2.2.1. Introduction

This game will be used as a tool for providing the basic education to kids and making the learning experience enjoyable with the factor of immersion and interaction. While playing the game, the user will be asked to solve different educational activities in a fun way. The game will also have various difficulties level so that the activities can be made harder as per user requirement. I made this project by combining the game and education which will encourage the creative thinking in kids.
Game can be played on Oculus quest with hand-controllers (might work on all Oculus VR headset, haven’t tested). VR technology for this project, it will provide us with the benefit of immersion and interaction. As the studies have shown, the children can learn better with interaction. So, the project will prove very beneficial for everyone (kids, teachers, parents etc).

2.2.2 Existing System

Virtual Reality is getting more prominent day by day. It provides the user immersive experience and interaction with the virtually created world. VR is being used in many fields for example: military, Sport, Medical training , Education etc. But, there are no good VR educational game for games. So, by making this project, I think I’ll be contributing something to the community and it will be very beneficial.

2.2.3. Organizational Chart

Organizational chart will be very much supportive to get a better overview of the game’s areas and their decomposition into different activities.

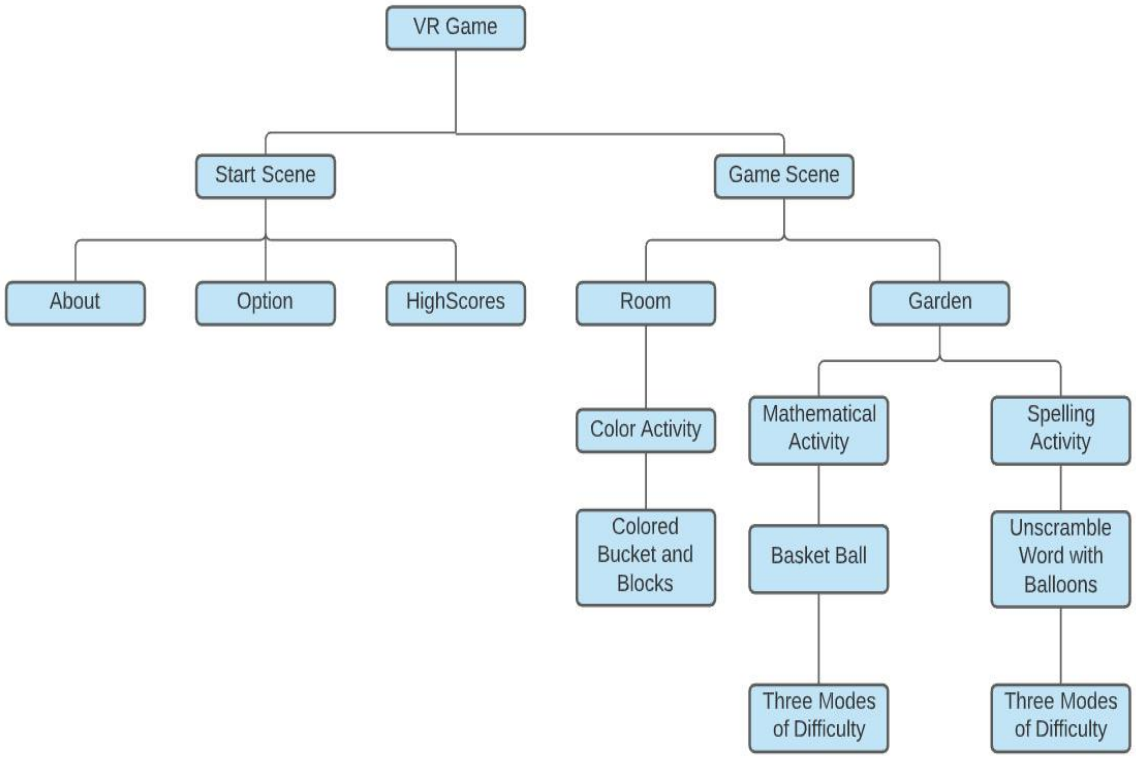


Figure 2.2: Organizational chart

2.2.4. Scope of the System

The Project will only support single player, there will be no 2-player or multiplayer support. As for now, project will not be released on any Online store, as it is a lengthy process and may require more time and quality assurance. The 3d assets, audio will not be designed by project creator as they are very time consuming and out scope for developer. The project will be created within the boundaries of coding, level designing, testing and logo designing. All the other mechanism like animation, sound effects, music , particle effects etc are included in level designing which will be implemented by developer.

2.2.5. Summary of Requirements: (Initial Requirements)

The purposed system must fulfill following requirements as follow:

Functional Requirements:

- To make the objects in environment interactable and apply proper physics on them. (For example user can hold and interact different objects and all the physics rule will Be applicable on object. For instance: if user throw object, object will have gravity).
- To provide soothing and immersive user experience. (Soothing because it is a game for kids. And immersive so that player can totally indulge into game).
- To implement mechanism of joints to doors and drawers. (So the doors and drawers can operate the way they operate in real world).
- To make player learn things by creating different fun activities and levels. (Learn alphabets, colors, spellings, mathematical operation and tables etc).
- To implement soft sound/audio mechanism. (Sound shouldn't be so loud).
- To implement animations where ever needed.

- To design fun theme for game.

Non-Functional Requirements:

- **Performance:** Optimization of game for better performance. The game shouldn't lag.
- **User Friendly:** Child-friendly UI and controls.
- **Portability:** The game will be developed for Oculus Quest, but can be ported to other VR headset with little bit of tweaking.
- **Good Graphics:**To get good graphics by customizing Post-Processing in unity.
- **Minimizing motion-sickness with better locomotion:** To minimize motion sickness. (motion sickness is one of the major drawback of VR. To make the game comfortable for kids, have to minimize motion sickness).

2.2.6. Identifying External Entities

As this is a game, there be only one external entity which is the player who will play this game. As the game is for kids, so the player more specifically speaking will be a **Kid**. Other than that, the device's memory can also be treat as an external entity, as the game will access memory to store and load the high score information.

2.2.7. Context Level Data Flow Diagram

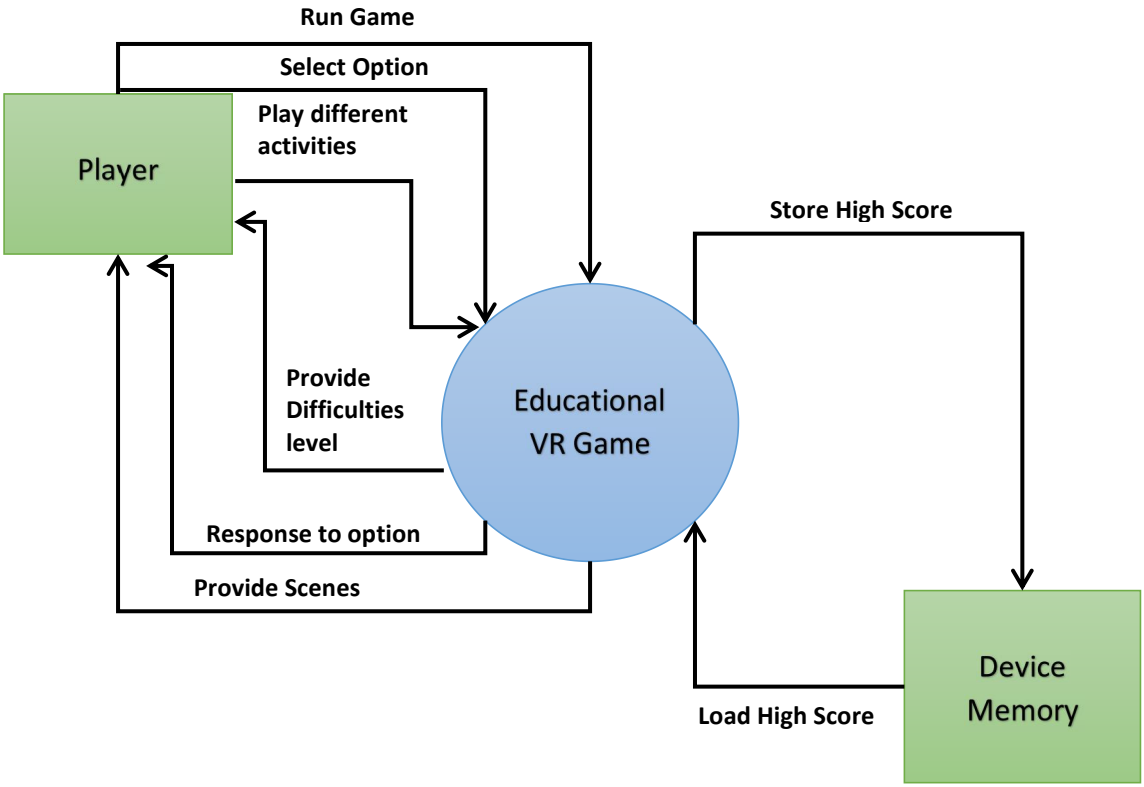


Figure 2.3: Context Level Data Flow Diagram

2.2.8. Capture "Shall" statements:

Table 2.1: Shall Statements (initial requirements)

No #	Initial Requirements
1.0	The environment “shall” be interactive containing interactable objects.
1.0	The user experience “shall” be immersive.
1.0	The user experience “shall” be pleasant.
1.0	The VR controllers “shall” provide haptic feedback.
2.0	The game “shall” contain sound effects (sfx).
2.0	The game “shall” contain background music.
2.0	The game “shall” contain visual affects (vfx).
2.0	The game “shall” contain animations.
2.0	The game “shall” be fun themed.
2.0	The game “shall” provide save mechanism.
2.0	The game “shall” provide load mechanism.
2.0	The game “shall” encourage creative thinking.
2.0	The game “shall” give positive and negative feedback while playing game.
2.0	The game “shall” have teleportation locomotion.
2.0	The game “shall” have continuous locomotion.
2.0	The game “shall” have hand grab mechanism.
2.0	The game “shall” have hand throw mechanism.
3.0	The player “shall” be given difficulties level to choose from.
3.0	The player “shall” learn mathematical operations while playing game.
3.0	The player “shall” learn spellings while playing game.
3.0	The player “shall” learn colors while playing game.

2.2.9. Allocate Requirements:

Table 2.2: Initial requirements and Use Case name

No #	Initial Requirements	Use Case Name
1.0	The environment "shall" be interactive containing interactable objects.	UC_interactive
1.0	The user experience "shall" be immersive.	UC_immersive
1.0	The user experience "shall" be pleasant.	UC_pleasant
1.0	The VR controllers "shall" provide haptic feedback.	UC_haptic_feedback
2.0	The game "shall" contain sound effects (sfx).	UC_sfx
2.0	The game "shall" contain background music.	UC_music
2.0	The game "shall" contain visual affects (vfx).	UC_vfx
2.0	The game "shall" contain animations.	UC_animation
2.0	The game "shall" be fun themed.	UC_theme
2.0	The game "shall" provide save mechanism	UC_save
2.0	The game "shall" provide load mechanism	UC_load
2.0	The game "shall" encourage creative thinking.	UC_creative
2.0	The game "shall" give positive and negative feedback while playing game.	UC_feedback
2.0	The game "shall" have teleportation locomotion.	UC_teleportation
2.0	The game "shall" have continuous locomotion.	UC_continuous_loco
2.0	The game "shall" have hand grab mechanism.	UC_grab
2.0	The game "shall" have hand throw mechanism.	UC_throw
3.0	The player "shall" be given difficulties level to choose from.	UC_difficulty_level
3.0	The player "shall" learn mathematical operations while playing game.	UC_maths
3.0	The player "shall" learn spellings while playing game.	UC_spellings
3.0	The player "shall" learn colors while playing game.	UC_colors

2.2.10. Prioritize Requirements:

Table 2.3: Prioritize requirements

No #	Rank	Initial Requirements	Use case id	Use Case Name
1.0	Highest	The environment "shall" be interactive containing interactable objects.	UC_1	UC_interactive
1.0	Highest	The user experience "shall" be immersive.	UC_2	UC_immersive
1.0	Medium	The user experience "shall" be pleasant.	UC_3	UC_pleasant
1.0	lowest	The VR controllers "shall" provide haptic feedback.	UC_4	UC_haptic_feedback
2.0	Medium	The game "shall" contain sound effects (sfx).	UC_5	UC_sfx
2.0	lowest	The game "shall" contain background music.	UC_6	UC_music
2.0	lowest	The game "shall" contain visual affects (vfx).	UC_7	UC_vfx
2.0	lowest	The game "shall" contain animations.	UC_8	UC_animation
2.0	Medium	The game "shall" be fun themed.	UC_9	UC_theme
2.0	lowest	The game "shall" provide save mechanism	UC_10	UC_save
2.0	lowest	The game "shall" provide load mechanism	UC_11	UC_load
2.0	Medium	The game "shall" encourage creative thinking.	UC_12	UC_creative
2.0	Medium	The game "shall" give positive and negative feedback while playing game.	UC_13	UC_feedback
2.0	Highest	The game "shall" have teleportation locomotion.	UC_14	UC_teleportation
2.0	lowest	The game "shall" have continuous locomotion.	UC_15	UC_continuous_loco
2.0	Highest	The game "shall" have hand grab mechanism.	UC_16	UC_grab
2.0	Highest	The game "shall" have hand throw mechanism.	UC_17	UC_throw
3.0	Medium	The player "shall" be given difficulties level to choose from.	UC_18	UC_difficulty_level
3.0	Highest	The player "shall" learn mathematical operations while playing game.	UC_19	UC_maths
3.0	Highest	The player "shall" learn spellings while playing game.	UC_20	UC_spellings
3.0	Medium	The player "shall" learn colors while playing game.	UC_21	UC_colors

2.2.11. Requirements Traceability Matrix:

Table 2.4: Traceability Matrix

No #	Initial Requirements	Build	Use - Case Name	Category
1.0	The environment "shall" be interactive containing interactable objects.	B1	UC_interactive	Business
1.0	The user experience "shall" be immersive.	B1	UC_immersive	Business
1.0	The user experience "shall" be pleasant.	B1	UC_pleasant	Business
1.0	The VR controllers "shall" provide haptic feedback.	B1	UC_haptic_feedback	Business
2.0	The game "shall" contain sound effects (sfx).	B1	UC_sfx	Business
2.0	The game "shall" contain background music.	B1	UC_music	Business
2.0	The game "shall" contain visual affects (vfx).	B1	UC_vfx	Business
2.0	The game "shall" contain animations.	B1	UC_animation	Business
2.0	The game "shall" be fun themed.	B1	UC_theme	Business
2.0	The game "shall" provide save mechanism	B1	UC_save	Business
2.0	The game "shall" provide load mechanism	B1	UC_load	Business
2.0	The game "shall" encourage creative thinking.	B1	UC_creative	Business
2.0	The game "shall" give positive and negative feedback while playing game.	B1	UC_feedback	Business
2.0	The game "shall" have teleportation locomotion.	B1	UC_teleportation	Business
2.0	The game "shall" have continuous locomotion.	B1	UC_continuous_loco	Business
2.0	The game "shall" have hand grab mechanism.	B1	UC_grab	Business
2.0	The game "shall" have hand throw mechanism.	B1	UC_throw	Business
3.0	The player "shall" be given difficulties level to choose from.	B1	UC_difficulty_level	Business
3.0	The player "shall" learn mathematical operations while playing game.	B1	UC_maths	Business
3.0	The player "shall" learn spellings while playing game.	B1	UC_spellings	Business
3.0	The player "shall" learn colors while playing game.	B1	UC_colors	Business

2.2.12. High Level Use case Diagram

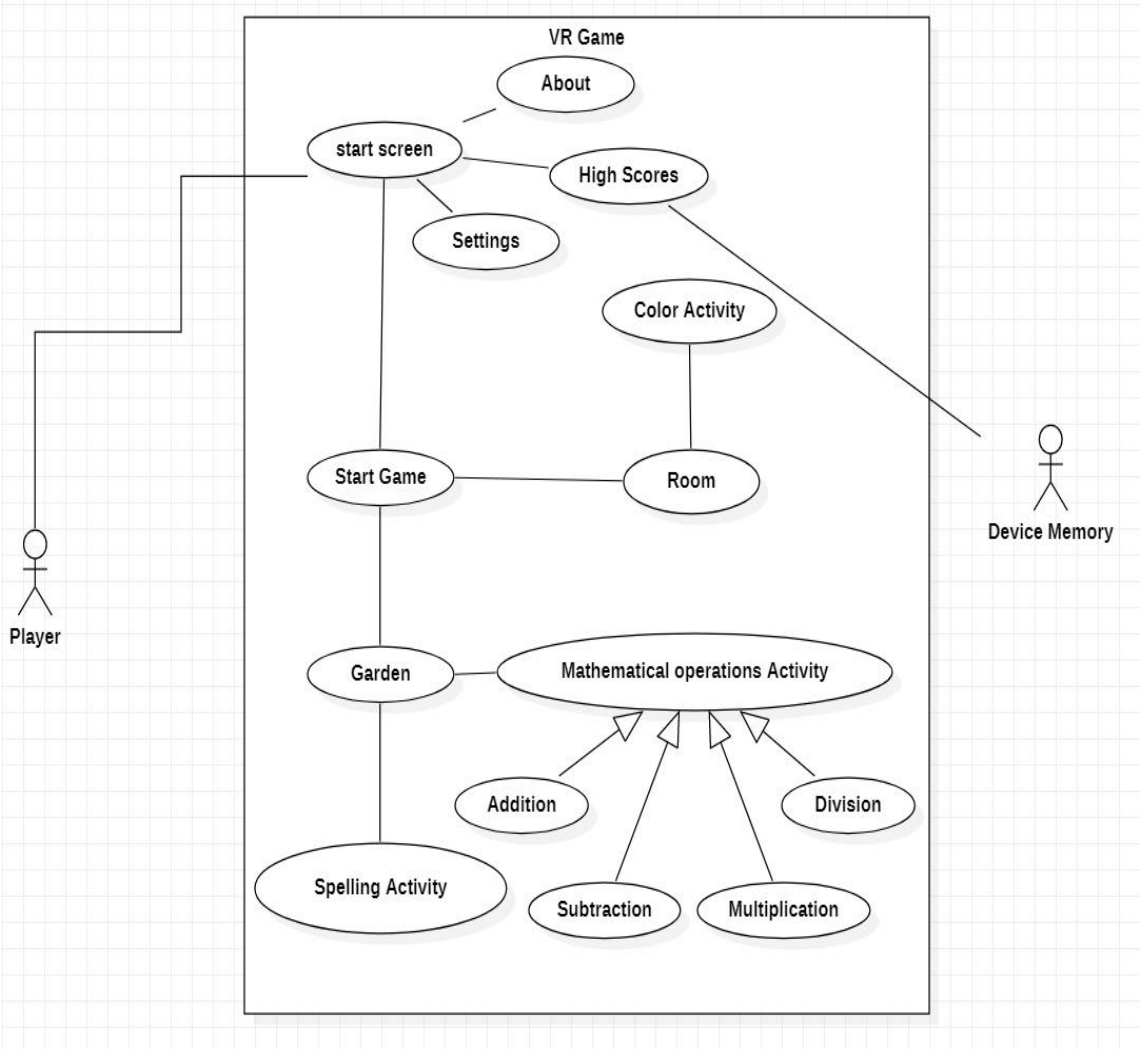


Figure 2.4: High Level Use case

2.2.13. Analysis Level Use case Diagram:

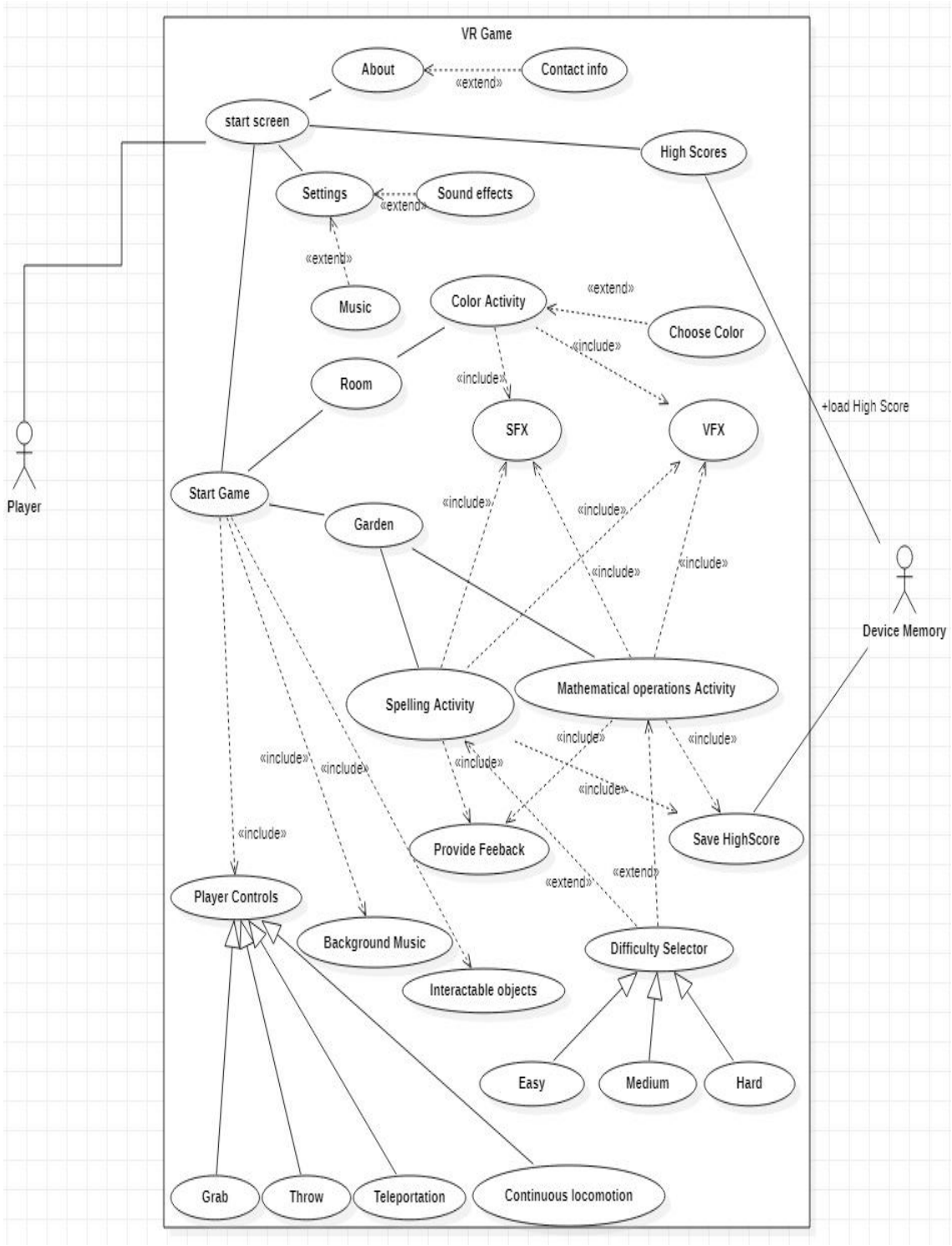


Figure 2.5: Analysis Level Use case

2.2.14. Use case Description:

Table 2.5: Use case description (interactive)

UC-ID	01
Use Case Name	UC_interactive
Actor	Player
Description	Player should be able to interact with objects in environment like drawer, door etc.
Precondition	1. Player have started the game. 2. VR headset and controllers are working fine.
Post-Condition	Player is able to interact with things.
Basic Flow	Player starts the game and interacts with object in environment. Player grab and throw things.
Alternative Flow	1. VR Headset and controllers are not working properly. 2. Reset Controllers. 3. Start Game. 4. The Object to interact with is not reachable. 5. Try Again by going closer.
Exception	1. Collision Error due to wrong collider. 2. Collider not scaled properly.

Table 2.6: Use case description (pleasant)

UC-ID	02
Use Case Name	UC_pleasant
Actor	Player
Description	Player should find the game environment pleasant and soothing.
Precondition	Player have started the game.
Post-Condition	Player finds the game environment pleasant.
Basic Flow	Player starts the game and observe the environment. They find the environment pleasant and easy on eye.
Alternative Flow	1. Player starts the game 2. Player finds the environment very unpleasant and unbearable. 3. Player closes the game.
Exception	1. Some objects in the environment loses color due to incompatible material. 2. Some player might find the colors too sharp .

Table 2.7: Use case description (haptic feedback)

UC-ID	03
Use Case Name	UC_haptic_feedback
Actor	Player
Description	Player should feel feedback on hands in the form of vibration while interacting with things.
Precondition	1. The hand controllers and their sensors are working properly. 2. The player should interact with any object.
Post-Condition	The player feels vibration on their hands while interacting with objects.
Basic Flow	The player starts the game and tries to interact with objects by grabbing the objects. While grabbing the object, they feel sensory feedback in the form of vibration on their hands.
Alternative Flow	1. The player starts the game and tries to grab object. 2. While grabbing, they feel no sensory feedback. 3. Player then, restarts game while resetting the controllers. 4. The haptic feedback is very little, so player couldn't feel it.
Exception	1. The haptic feedback is too strong that it scares the player. 2. The haptic feedback is too weak that player couldn't feel it.

Table 2.8: Use case description (Sound Effects)

UC-ID	04
Use Case Name	UC_sfx
Actor	Player
Description	The sound effects (sfx) should play to give the appropriate feedback to user in form of sound (for example, ball collision sound).
Precondition	1. The VR headset's speaker should be working fine. 2. To listen the sfx, player should perform or play any activity. (throw ball).
Post-Condition	The player listens the sounds while playing activity and this makes the experience more real and pleasant.
Basic Flow	Player starts the game and plays any activity.While playing the activity, Player hear a sound. The sound also plays when player provides an answer or interacts with objects during activity.
Alternative Flow	1. The Player Starts activity, but couldn't hear sound effect while interacting with objects. 2. Player volume up the sound effects. 3. Player check the speaker of VR headset and replay the game.
Exception	1. SFX are too low that it makes them inaudible. 2. SFX are too loud. 3. SFX might not play due to internal logic of program. 4. SFX file is corrupted.

Table 2.9: Use case description (Visual Effects)

UC-ID	05
Use Case Name	UC_vfx
Actor	Player
Description	The visual effects (vfx) should play to give the appropriate feedback to user in form of visuals (for example, the fireworks on correct guess).
Precondition	To see the vfx, player should perform or play any activity. (make a correct guess, throw ball etc).
Post-Condition	The player sees the visuals while playing activity. The player is awestruck seeing the vfx which then encourage him to answer more accurately.
Basic Flow	Player starts the game and plays any activity.While playing the activity, Player sees a visual effect. The vfx also plays when player provides an answer or interacts with objects during activity.
Alternative Flow	1. The Player Starts activity, but couldn't see any vfx. 2. Vfx might not be pleasant. 3. Vfx might not be visible due to poor
Exception	1. VFX file is corrupted. 2. VFX couldn't play due to the internal logic of game.

Table 2.10: Use case description (save)

UC-ID	06
Use Case Name	UC_save
Actor	Player, device memory
Description	The score of activity gets save and stores in device memory whenever player makes high-score.
Precondition	The player should be playing activity and able to make high-score.
Post-Condition	Player makes high score and it gets stored on device memory.
Basic Flow	Player starts the game and play any activity of their choice. When player scores highest, it gets stored in device memory.
Alternative Flow	1. Player plays an activity and make high score but it doesn't get stored in device memory. 2. The game still shows the previous high score.
Exception	1. Insufficient device memory. 2. File handling error.

Table 2.11: Use case description (load)

UC-ID	07
Use Case Name	UC_load
Actor	Player, device memory
Description	When player starts the game, the high-score gets automatically loaded into game.
Precondition	There should be already saved high-score file.
Post-Condition	The scores are loaded correctly and are visible to player.
Basic Flow	Player starts the game and go to main menu's "High-score" option. There, they could see the high-score opposite to each activity. Player plays the game, the previously scored high-score gets loaded and over-written if player score higher.
Alternative Flow	1. The player starts the game and goes to main menu's "High-score" but there are no high -score available. 2. Previously, stored high-score gets delete. 3. Wrong high-scores are shown.
Exception	1. File-handling error. 2. Incorrect high-scores, due to internal game logic.

Table 2.12: Use case description (teleportation)

UC-ID	08
Use Case Name	UC_teleportation
Actor	Player
Description	The player should be able to teleport from one place to another using VR controller.
Precondition	The game should be started and controllers should be working properly. The targeted place to teleport should be accessible and should have no obstacles.
Post-Condition	The player teleports to targeted area.
Basic Flow	The player starts the game. Then aims the teleportation line on the accessible area. By pressing the button on controller , player gets teleport to the desire destination.
Alternative Flow	1. The player starts the game and aims at the location to teleport. 2. The location is inaccessible due to the problem in "navmesh" settings. 3. The teleportation button on controller is not working.
Exception	1. Problem in NavMesh settings. 2. Internal logic error.

Table 2.13: Use case description (grab)

UC-ID	09
Use Case Name	UC_grab
Actor	Player
Description	The player should be able to grab the objects in game. The grabbing action will be performed with hand controllers, using grab button. The sensitivity of grab button will also affect the mechanism.
Precondition	The controllers should be working properly. The player should know the grab button.
Post-Condition	The targeted object will be held into the hand of player as long as player held the grab button down.
Basic Flow	The player will start the game and will take their hand towards the target object and press grab button.
Alternative Flow	The player take their hands towards target object but couldn't hold it due to the wrong button press.
Exception	1. The grab button's sensitivity not working accurately. 2. The targeted object doesn't have "grab-able" component on it. 3. The targeted object doesn't have collider. 4. The targeted object's collider is not set correctly. 5. The collision between hand and object is not detected.

Table 2.14: Use case description (difficulty level)

UC-ID	10
Use Case Name	UC_difficulty_level
Actor	Player
Description	The player should be given choice to select the difficulty level of the activities. There should be three levels : Easy, Medium and Hard. The easy level should increment score with small margin, medium should increase with medium margin and hard mode should increase the scores with highest margin.
Precondition	The player is in activity area. The player is interacting with difficulty board by standing in front of it.
Post-Condition	The player selects the desired difficulty level and starts the activity.
Basic Flow	1. Player goes to any activity area in garden. 2. Player stands in front of interactive difficulty board. 3. Player moves the hand to the desired difficulty level and press the select button. 4. The activity gets easier or harder according to the choose option.
Alternative Flow	1. The player is not able to interact with difficulty board. 2. The difficulty level doesn't get applied on activity. 3. The scores are not getting updated according to difficulty.
Exception	1. The difficulty board "canvas" is not set properly. 2. The difficulty level board's button are not scaled properly. 3. The collision detection error.

Table 2.15: Use case description (maths)

UC-ID	11
Use Case Name	UC_maths
Actor	Player
Description	The Player should be able to learn mathematical operations while playing the maths activity. It should be designed in such way that player gets motivated and tries to solve the maths problem eagerly. The player should also have fun while playing it. It should support : addition, subtraction, multiplication and division.
Precondition	The player should know a little bit about mathematical operations. The hand controllers and VR headset should be working properly. The player should be able to read problems.
Post-Condition	The player answer the problem correctly and is able to make their maths skill sharper. Meanwhile, they also enjoyed while learning.
Basic Flow	1. The player will start the game and goes to the maths section in garden. 2. The player starts the activity by pressing the button. 3. The player reads the question on the board and tries to solve it by providing a correct answer. (i.e. by scoring the ball in correct basket)
Alternative Flow	1. The player starts the activity but activity doesn't get started. 2. The Ball doesn't get instantiated. 3. The baskets doesn't contain the correct answer. 4. The question is too easy or too hard for player that they get bored. 5. The player is not enjoying while playing the activity.
Exception	1. Null reference exception (if program tries to access something which has been destroyed) 2. The error while instantiating the ball. 3. The collision detection error. 4. Program's internal logic error.

Table 2.16: Use case description (spellings)

UC-ID	12
Use Case Name	UC_spellings
Actor	Player
Description	The Player should be able to learn spellings while playing the spelling activity. It should be designed in such way that player gets motivated and tries to solve it with eagerness. It should also encourage the creative thinking in player. The player should also have fun while playing it.
Precondition	The hand controllers and VR headset should be working properly. The player should be able to read problem.
Post-Condition	The player solve the problem correctly and is able to make their spelling skills sharper. Meanwhile, they also enjoyed while learning.
Basic Flow	1. The player will start the game and goes to the spelling section in garden. 2. The player starts the activity by pressing the button. 3. The player reads the scrambled spelling on the stage and tries to unscramble it. 4. Player unscrambles the spelling and their score gets increased.
Alternative Flow	1. The player starts the activity but activity doesn't get started. 2. The Balloons doesn't get instantiated. 3. The player unscramble the word correctly but the games give negative feedback. 4. The question is too easy or too hard for player that they get bored. 5. The player is not enjoying while playing the activity.
Exception	1. Null reference exception (if program tries to access something which has been destroyed) 2. The error while instantiating the balloon. 3. The collision detection error. 4. Program's internal logic error. 5. The searching algorithm not working properly. 6. The feedback from the game is given wrong.

Table 2.17: Use case description (colors)

UC-ID	13
Use Case Name	UC_colors
Actor	Player
Description	The Player should be able to learn colors while playing the color activity. It should be designed in such way that player gets motivated and tries to solve it with eagerness. It should also encourage the creative thinking in player. The player should also have fun while playing it. The player should also be given option to choose the colors, they want to learn.
Precondition	The hand controllers and VR headset should be working properly. The player should be able to read problem and understand it. (the guardian can also help in making them understand the problem). The player should be in room. (game's room)
Post-Condition	The player solve the problem correctly and is able to learn about colors. Meanwhile, they also enjoyed while learning. The baskets get destroy after player put the colored boxes in their respective baskets.
Basic Flow	1. The player will start the game and reads the sign board. 2. The player is not allowed to go outside in garden until they solve the problem. 3. The player selects any three colors and press okay to start the activity. 4. The player finds the colored boxes scattered in the room. 5. Player put the boxes in their respective colored baskets. 6. When all the boxes are collected and are put in their respective baskets, the baskets get disappear. 7. The room door gets unlock and player is allowed to go in the garden.
Alternative Flow	1. The player selects the colors and starts the activity. 2. The activity doesn't get started. 3. The player is not able to find the hidden boxes. 4. The room door remains locked and player gets bored and leave the game.
Exception	1. Null reference exception (if program tries to access something which has been destroyed) 2. The collision detection error. 3. Program's internal logic error. 4. The blocks and baskets not get destroyed correctly. 5. The feedback from the game is given wrong.

Chapter 3: Design Document (For Object Oriented Approach)

3.1. Introduction

Third deliverable is all about the software design. In the previous deliverable, analysis of the system is completed. So we understand the current situation of the problem domain. Now we are ready to strive for a solution for the problem domain by using object-oriented approach. Following artifacts must be included in the 3rd deliverable.

1. Domain Model
2. System Sequence Diagram
3. Sequence Diagram
4. Collaboration Diagram
5. Design Class Diagram
6. State Transition Diagram
7. Data Model

Now we discuss these artifacts one by one as follows:

3.2. Domain Model

Domain models represent the set of requirements that are common to systems within a product line. There may be many domains, or areas of expertise, represented in a single product line and a single domain may span multiple product lines. The requirements represented in a domain model include: Definition of scope for the domain, Information or objects, Features or use cases, including factors that lead to variation, Operational/behavioral characteristics

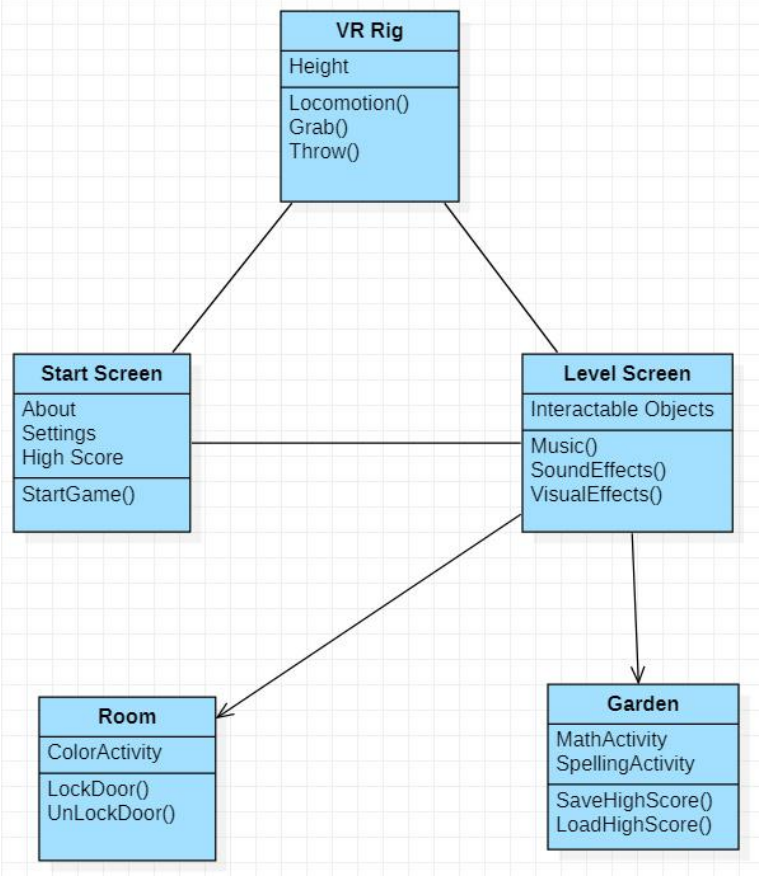


Figure 3.1: Domain Model

3.3. Sequence Diagram

A Sequence diagram depict the series of events that happen in a system. The incantation of method in each object, and the sort in which the incantation occur is captured in a Sequence diagram. This makes the Sequence diagram a very helpful means to easily signify the forceful activities of a system.

A Sequence diagram is two-dimensional in personality. On the straight axis, it shows the life of the thing that it represent, while on the perpendicular axis, it shows the sequence of the making or incantation of these objects.

Because it uses class given name and thing name references, the Sequence diagram is very useful in elaborate and detail the forceful drawing and the sequence and beginning of incantation of objects. Hence, the Sequence diagram is one of the most broadly used forceful diagrams in UML.

A sequence diagram is prepared up of objects and messages. substance are stand for accurately how they have been represent in all UML diagrams—as rectangles with the underscore class name within the rectangle.

Sequence diagrams explain associations between classes in fundamentals of an swap over of messages over time.

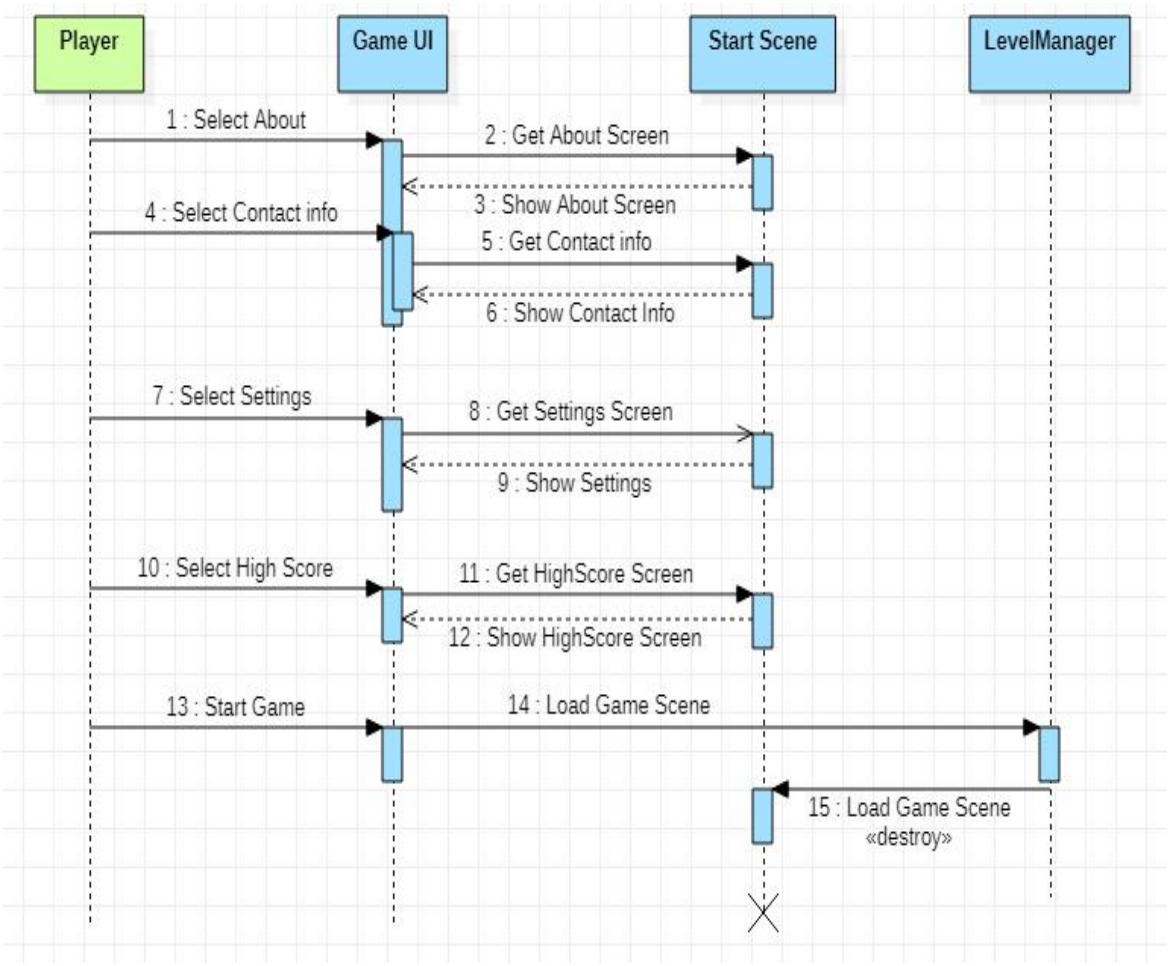


Figure 3.2: Game UI Sequence Diagram

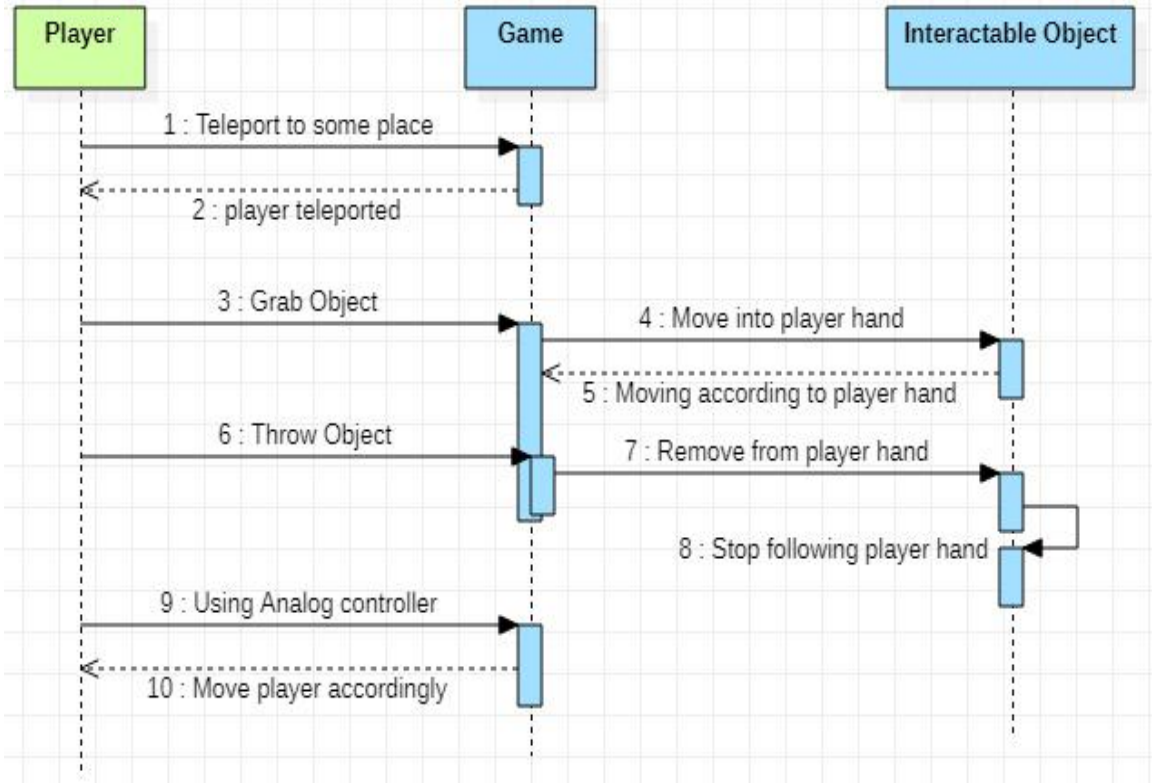


Figure 3.3: Interactable Objects Sequence diagram

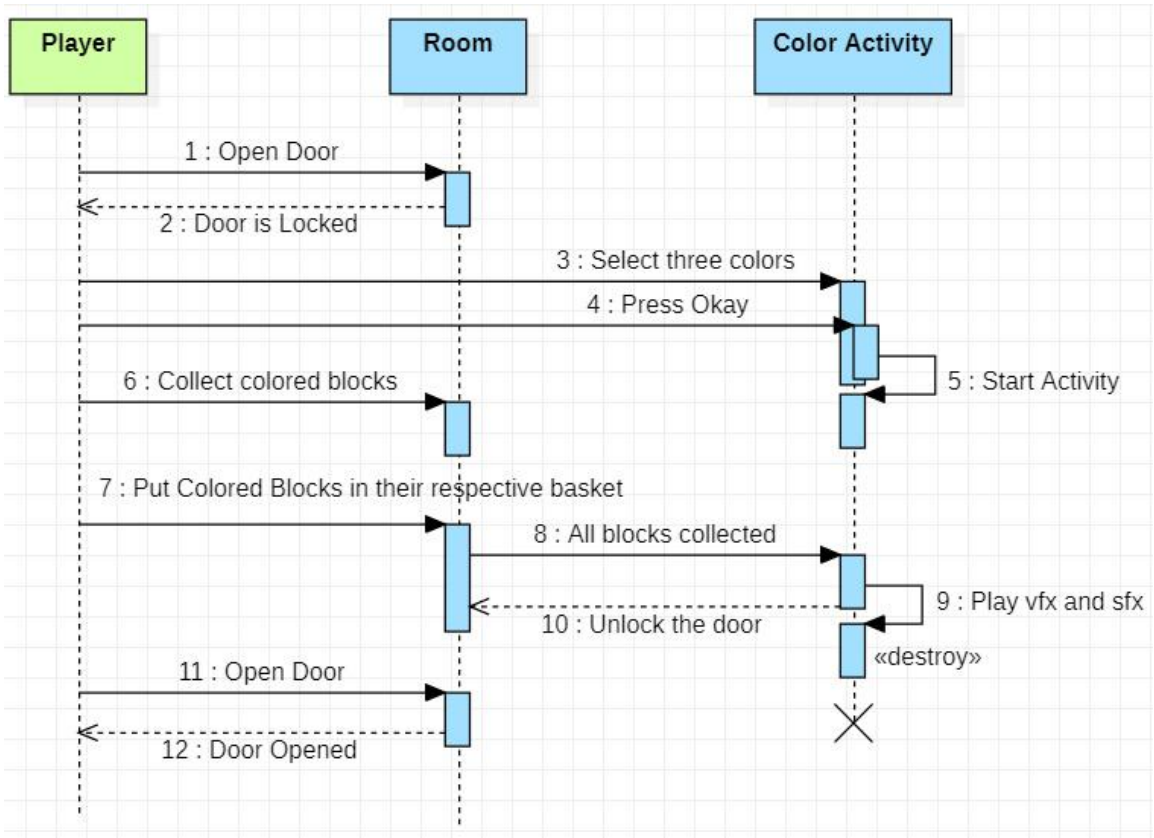


Figure 3.4: Color Activity Sequence diagram

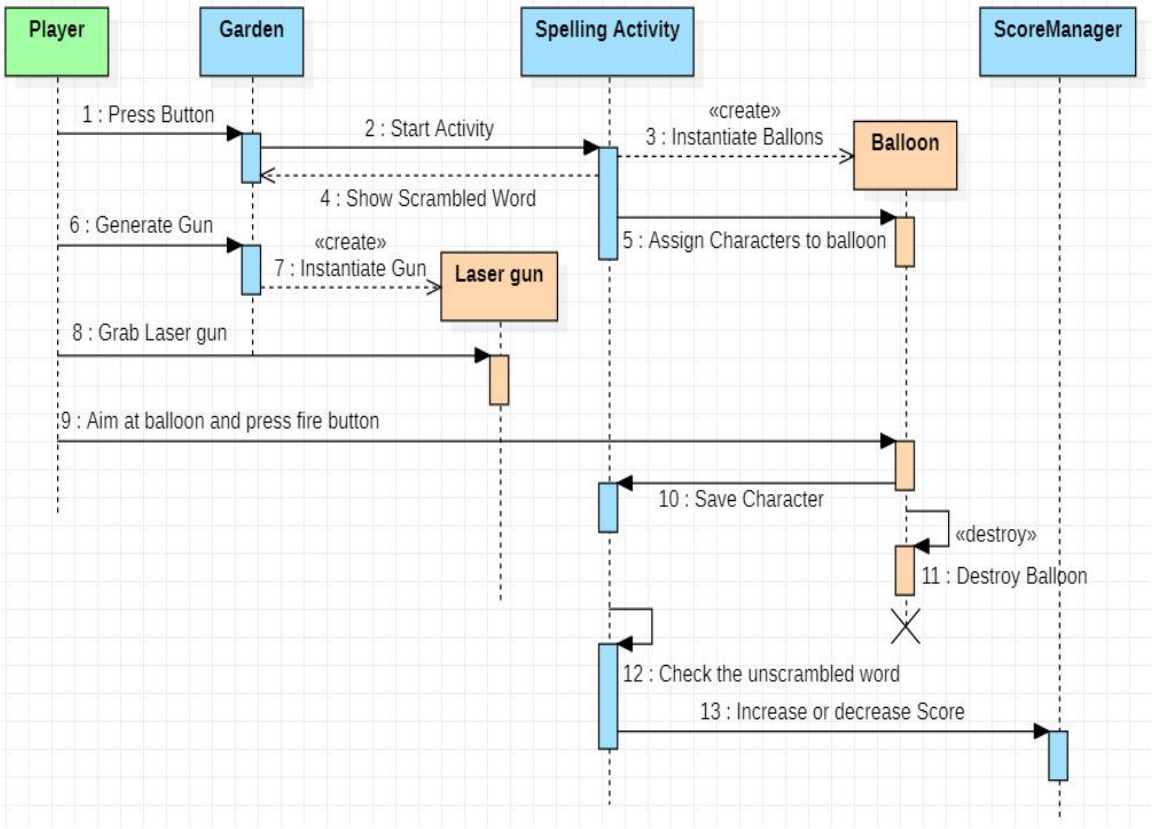


Figure 3.5: Spelling Activity Sequence diagram

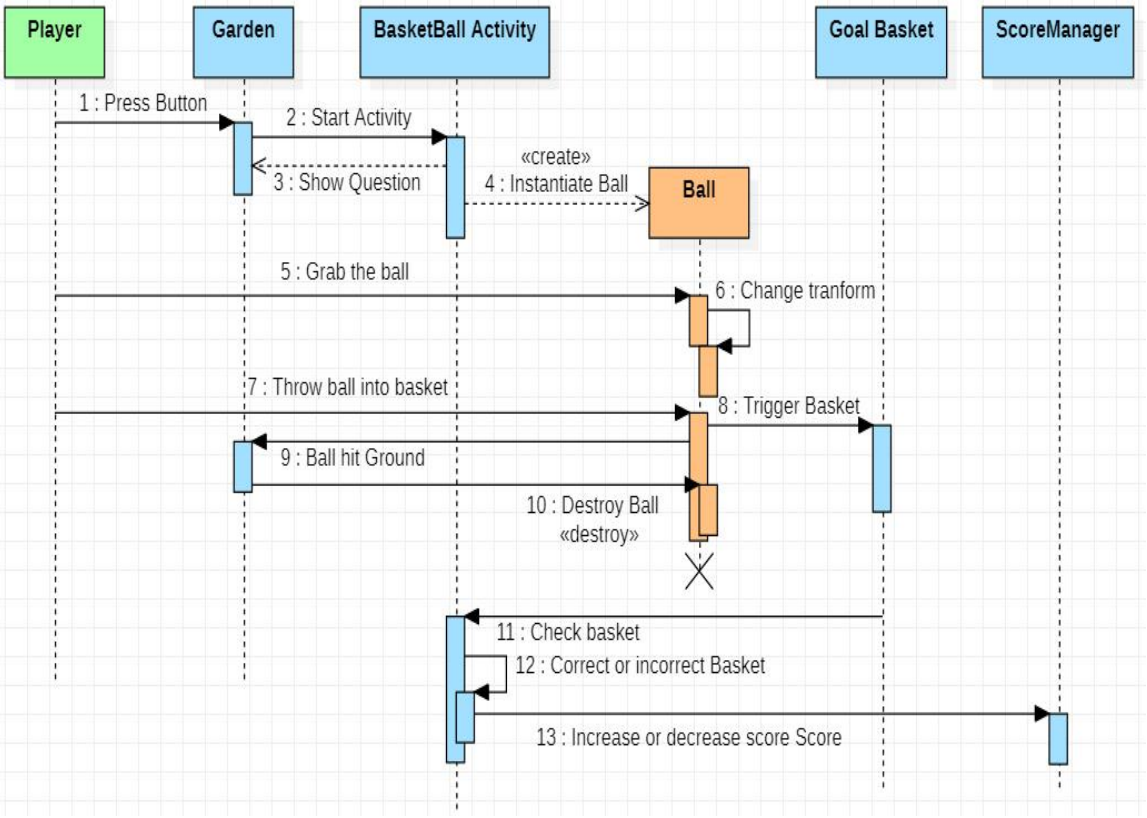


Figure 3.6: Basketball Activity Sequence diagram

3.4. Collaboration Diagram

A collaboration diagram describes a model of announcement among objects; it shows the objects partake in the boundary by their relations to each other and the messages that they send to each other.

Collaboration diagrams are used to show how objects interrelate to carry out the actions of a exacting use case, or a part of a use case. the length of with sequence diagrams, collaborations are used by designer to describe and explain the role of the objects that carry out a particular gush of events of a use case. They are the most important source of information used to decisive class everyday jobs and interfaces.

not like a sequence diagram, a collaboration diagram shows the associations among the objects. Sequence diagrams and collaboration diagrams articulate alike in sequence, but show it in different ways. Collaboration diagrams show the relations among objects and are improved for accepting all the property on a given object and for practical design.

Because of the arrangement of the collaboration diagram, they be apt to superior right for analysis activities. specially, they be likely to be superior right to depict simpler communications of smaller numbers of objects. As the number of objects and messages grow, the diagram become more and more hard to read. In adding, it is difficult to show supplementary expressive in sequence such as timing, decision points, or other shapeless information that can be easily new to the comments in a sequence diagram.

3.4.1. Player and Start-Scene Collaboration

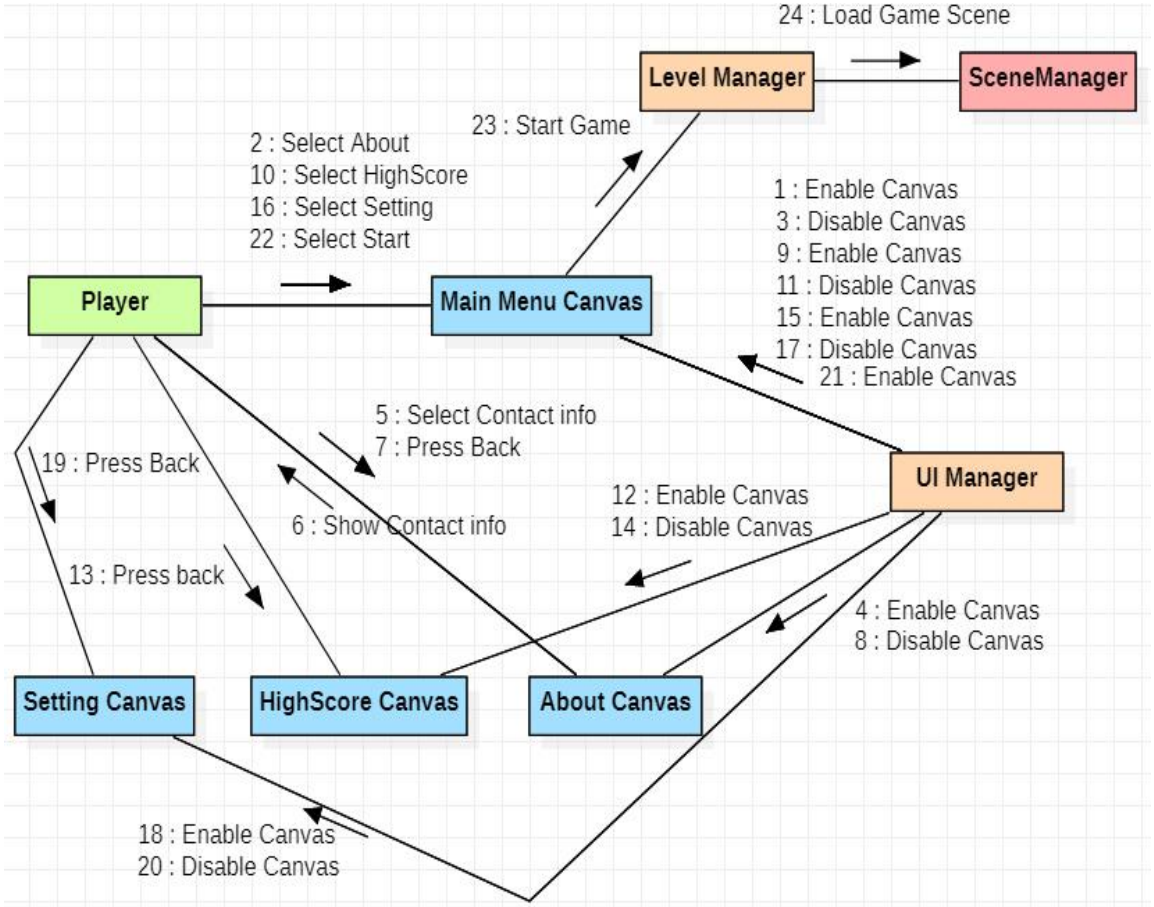


Figure 3.7: Player and Start-Scene Collaboration Diagram

3.4.2. Player and Basketball-Activity Collaboration

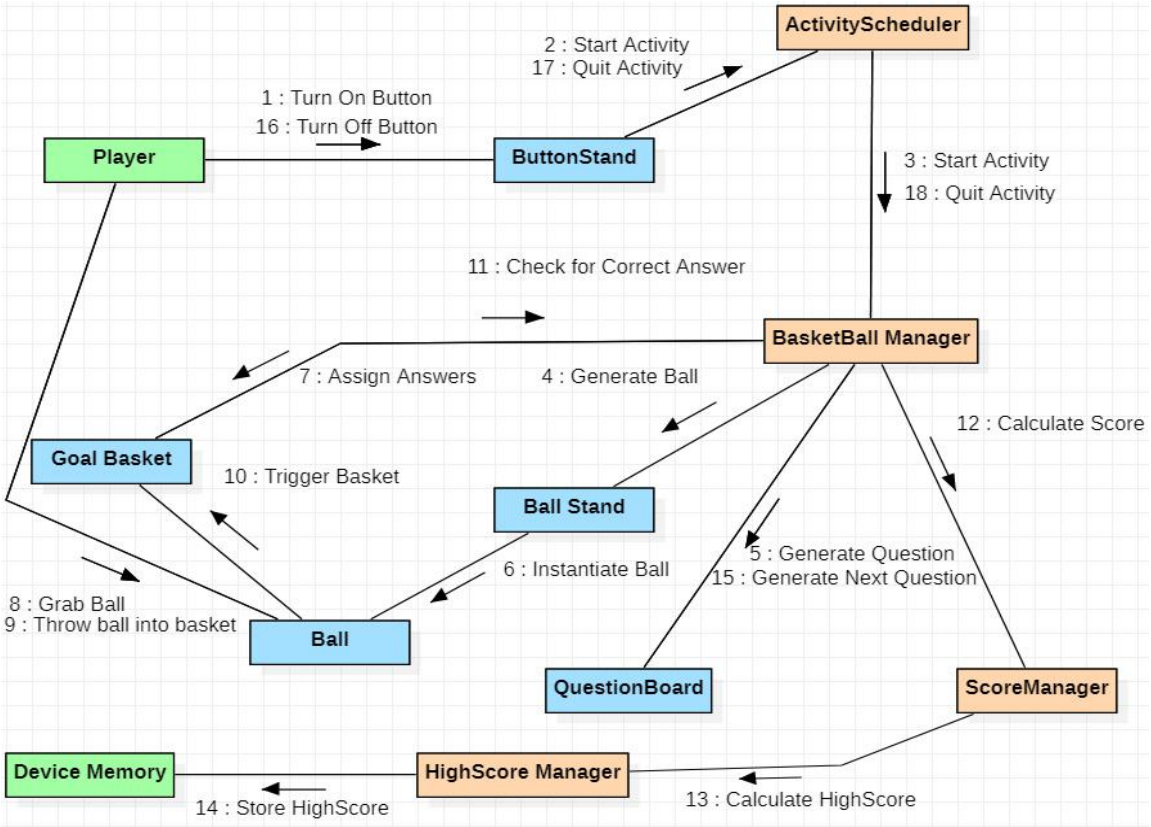


Figure 3.8: Player and Basketball-Activity Collaboration diagram

3.4.3. Player and Color-Activity Collaboration

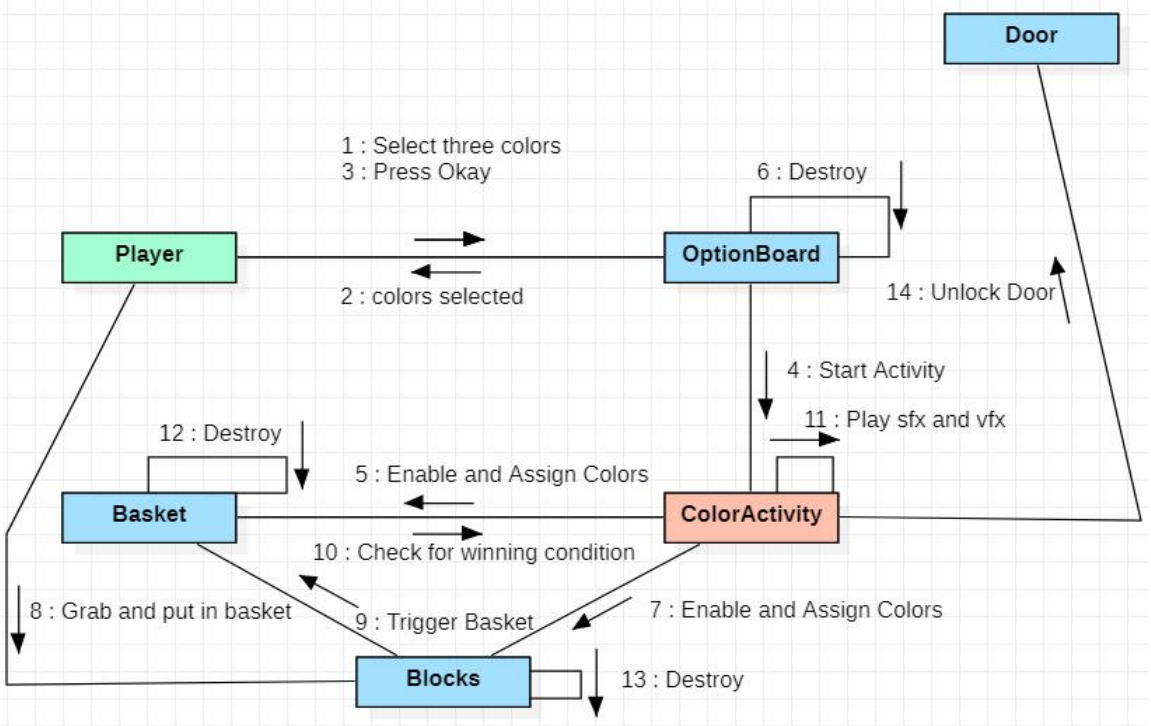


Figure 3.9: Player and Color-Activity Collaboration

3.4.4. Player and Spelling-Activity Collaboration

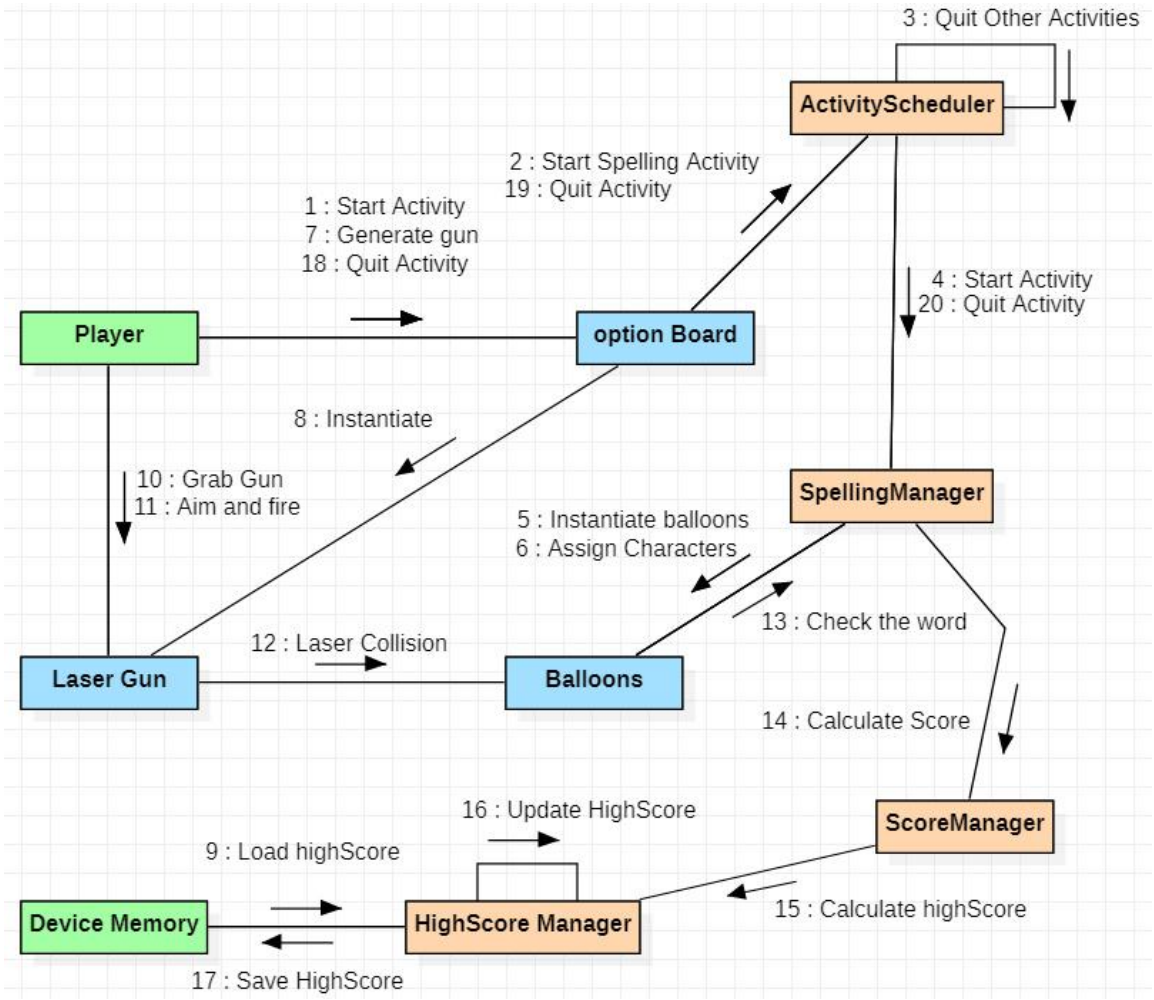


Figure 3.10: Player and Spelling-Activity Collaboration

3.5. Class Diagram

Classes are the work-horses of the design effort—they actually perform the real work of the system. The other design elements—subsystems, packages and collaborations simply describe how classes are grouped or how they interoperate.

3.5.1. Start Screen

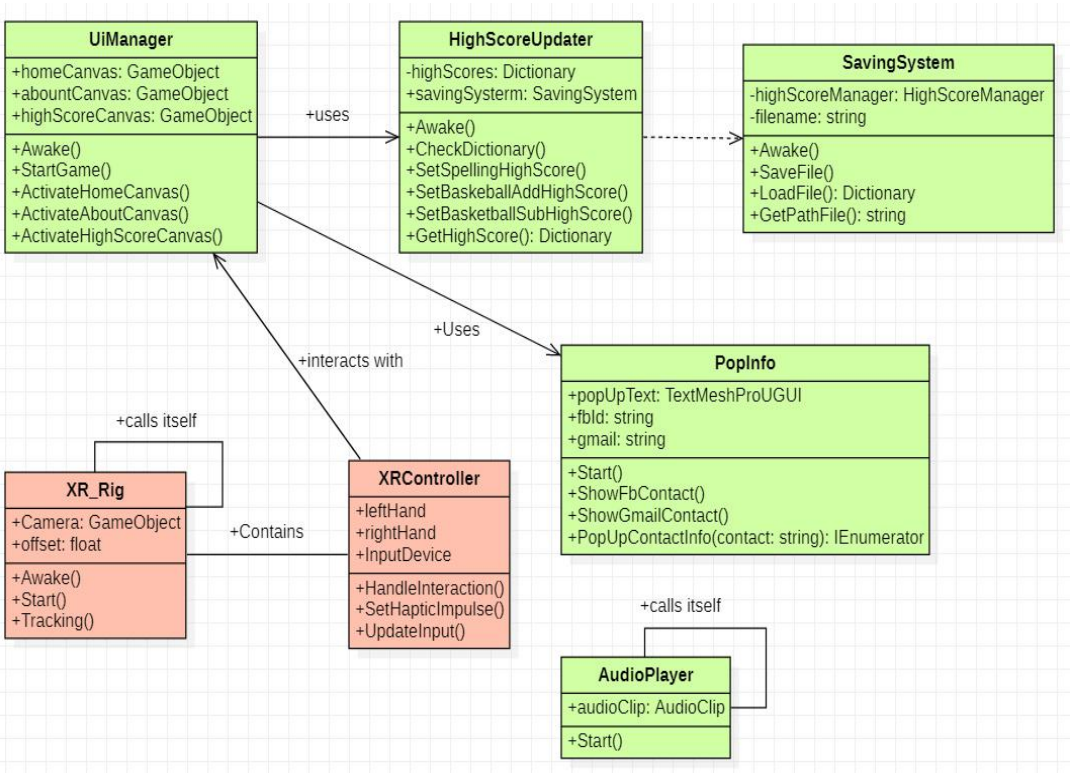


Figure 3.11: Start Screen Class Diagram

3.5.2. Color-Activity

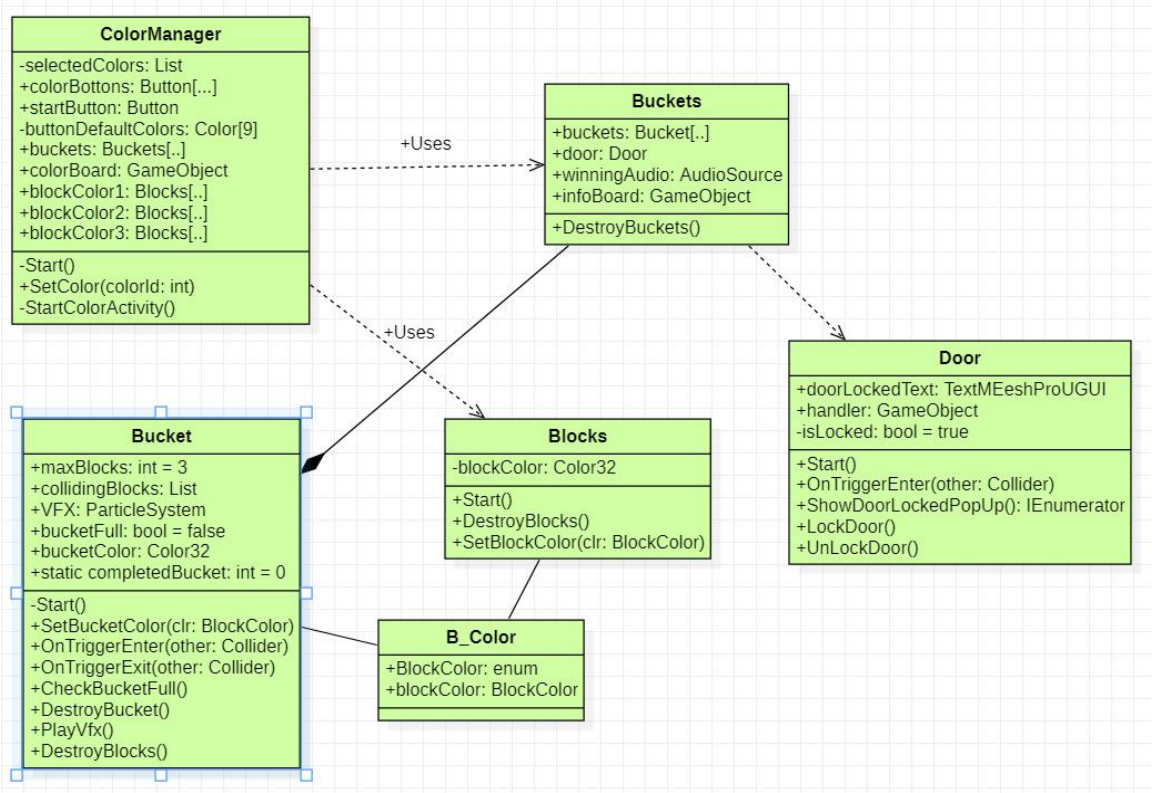


Figure 3.12: Color-activity Class Diagram

3.5.3. Basketball-Activity

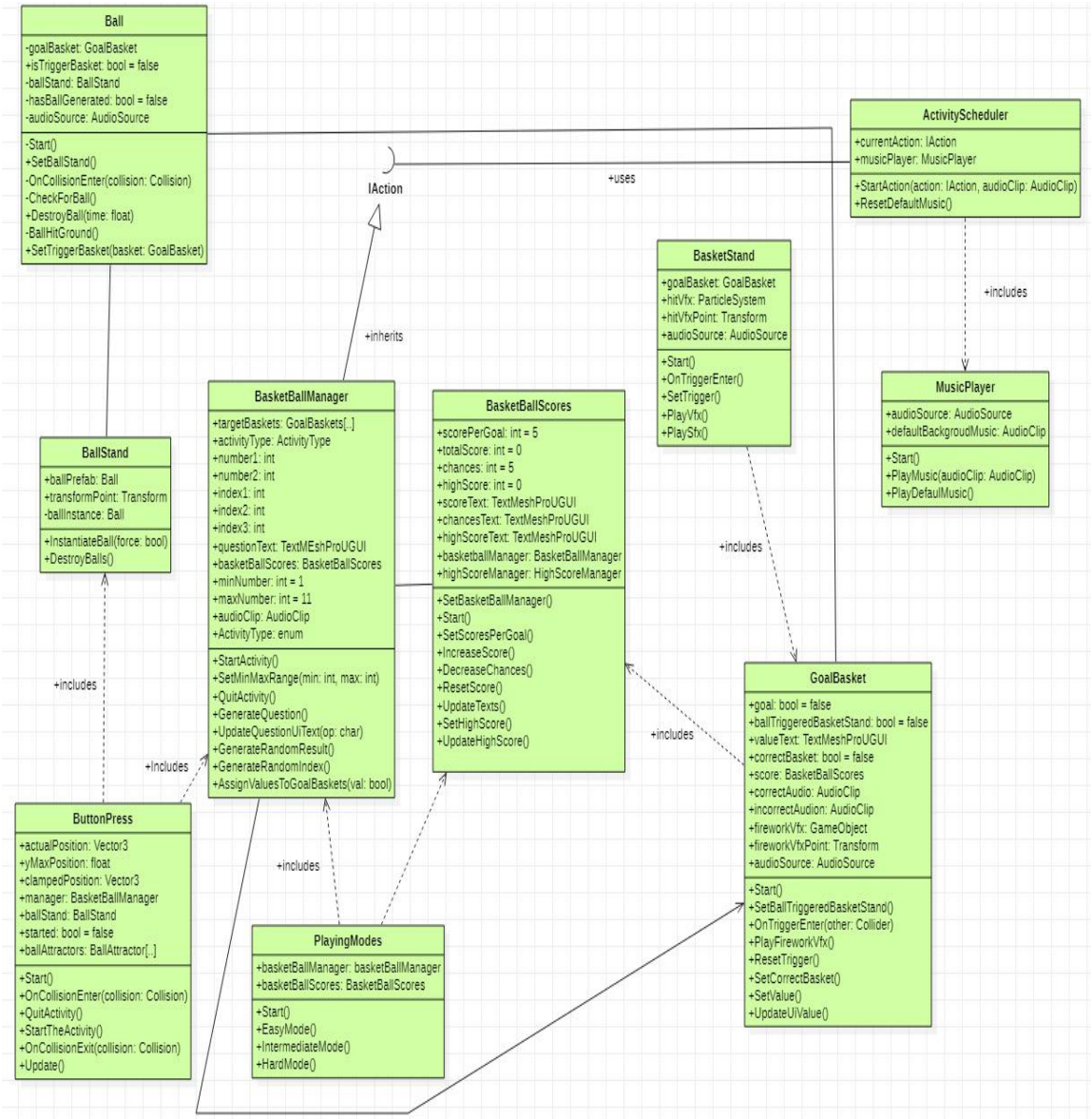


Figure 3.13: Basketball-activity Class Diagram

3.5.4. Spelling-Activity

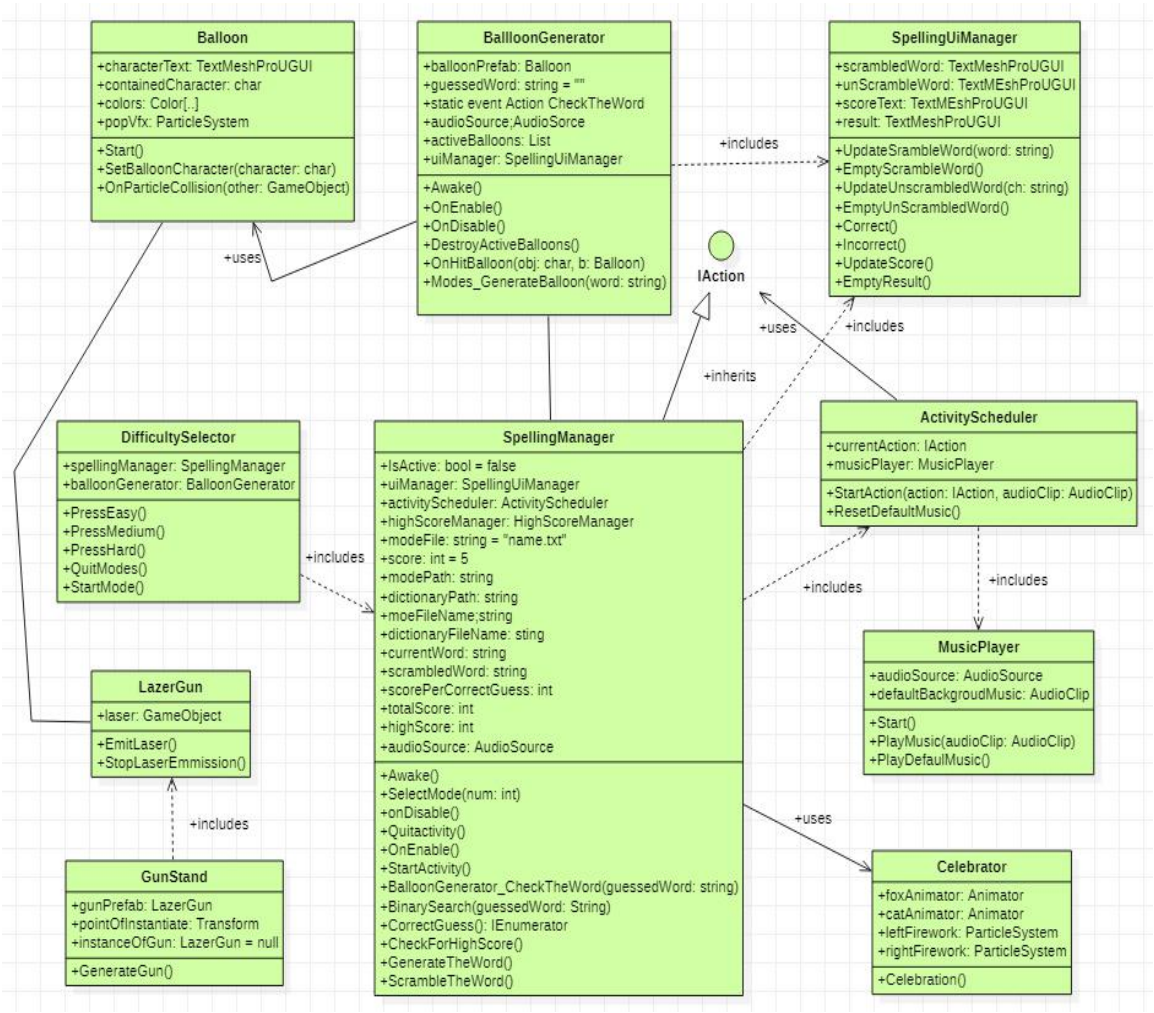


Figure 3.14: Spelling-activity Class Diagram

3.6. State chart Diagram

For some operations, the behavior of the operation depends upon the state the receiver object is in. A state machine is a tool for describing the states the object can assume and the events that cause the object to move from one state to another. State machines are most useful for describing active classes. The use of state machines is particularly important for defining the behavior.

3.6.1. Start Screen State chart diagram

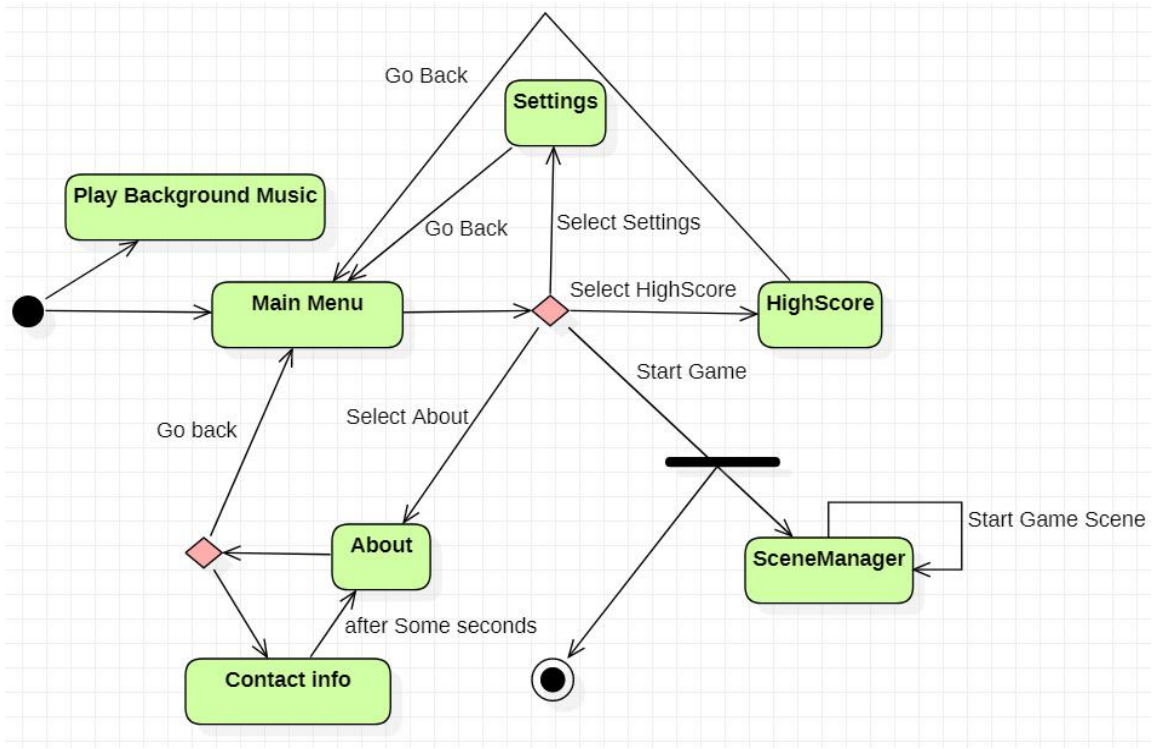


Figure 3.15: Start Screen State chart Diagram

3.6.2. Color-Activity State chart diagram

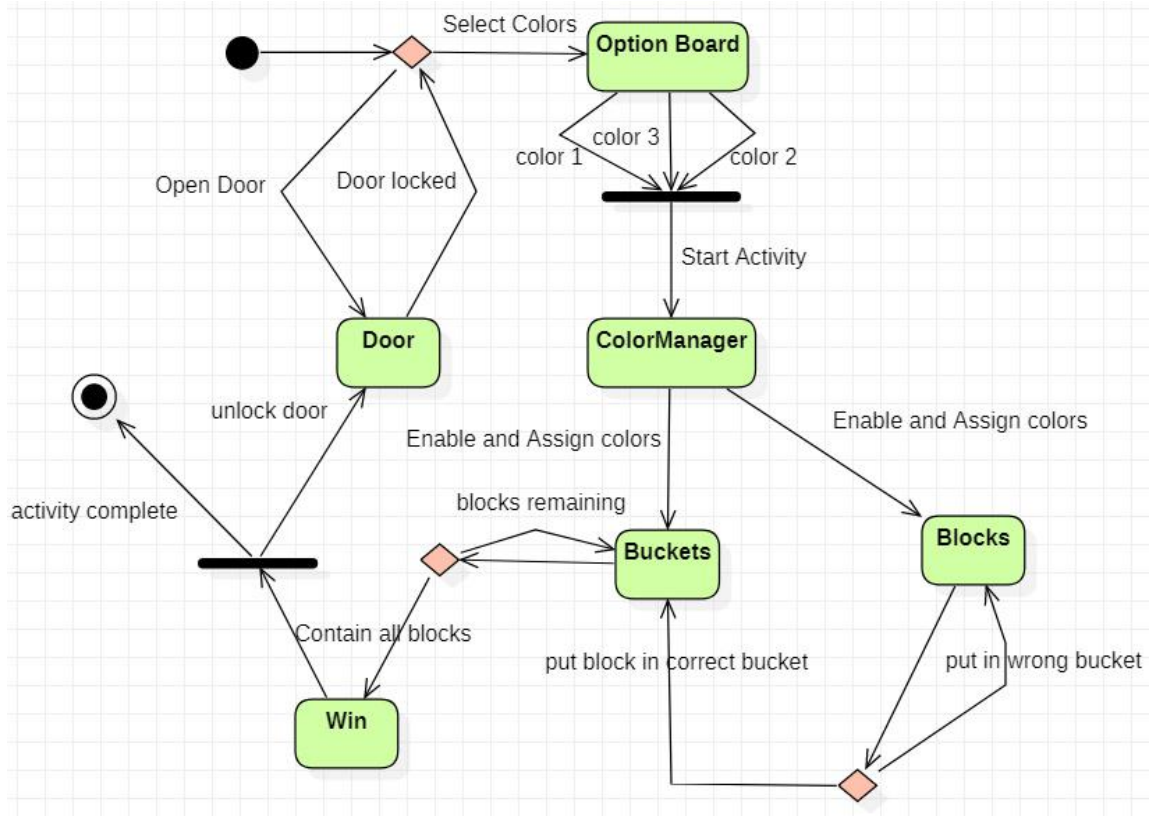


Figure 3.16: Color-activity State chart Diagram

3.6.3. Basketball-Activity State chart diagram

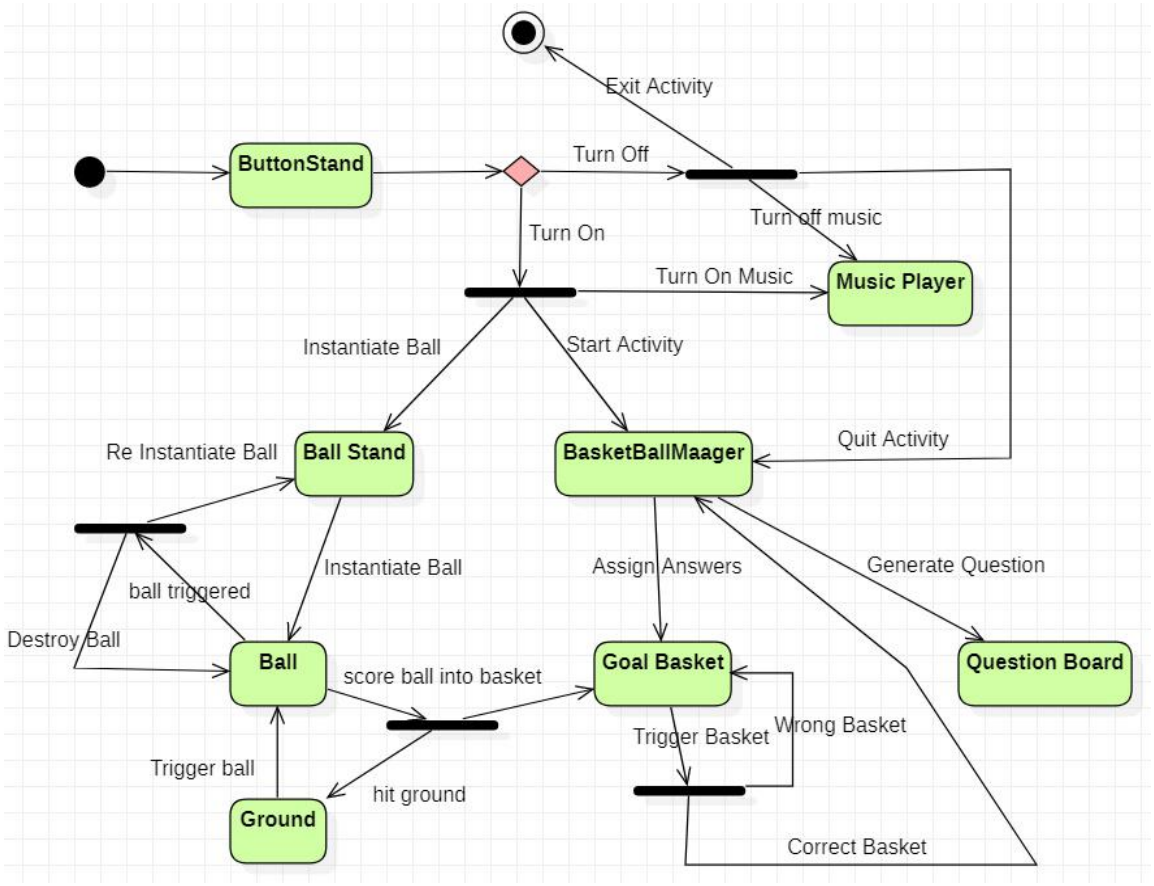


Figure 3.17: Basketball-activity State chart Diagram

Chapter 4: User Interface Design

4.1.Introduction

A user interface design consists of three main parts:
Page elements should be visualized on paper before building them in the computer. Just as you draw a site map to plan the site, use cartoons and storyboards to begin blocking out the site's appearance and navigational scheme.

1. Site maps
2. Storyboards
3. Navigational maps
4. Traceability Matrix

4.2. Site maps

A site map's main benefit is to give users an overview of the site's areas in a single glance by dedicating an entire page to a visualization of the information architecture. If designed well, this overview can include several levels of hierarchy, and yet not be so big that users lose their ability to grasp the map as a whole.

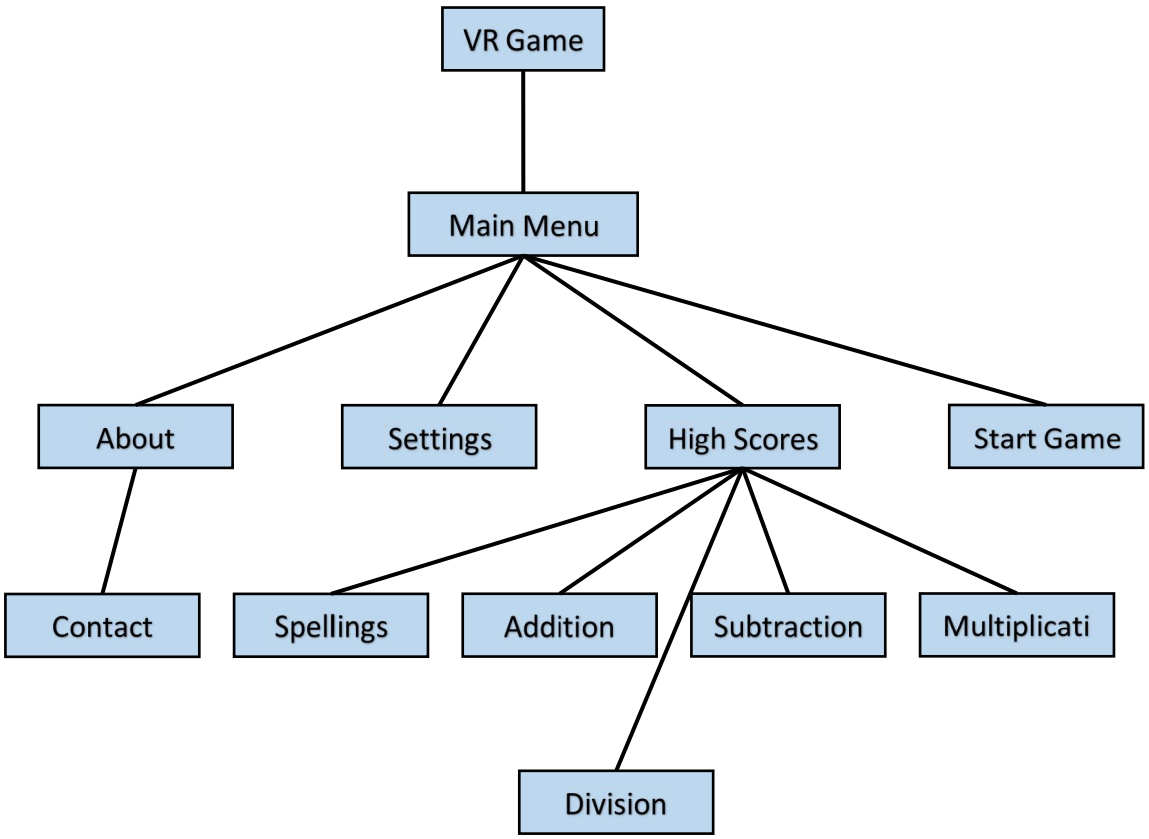


Figure 4.1: Site maps

4.3. Story boards

A storyboard is a sequence of single images, each of which represents a distinct event or narrative. It is also a visual representation of the script illustrating the interaction between the user and the machine. It can also be imagined as a film in visual-outline form.

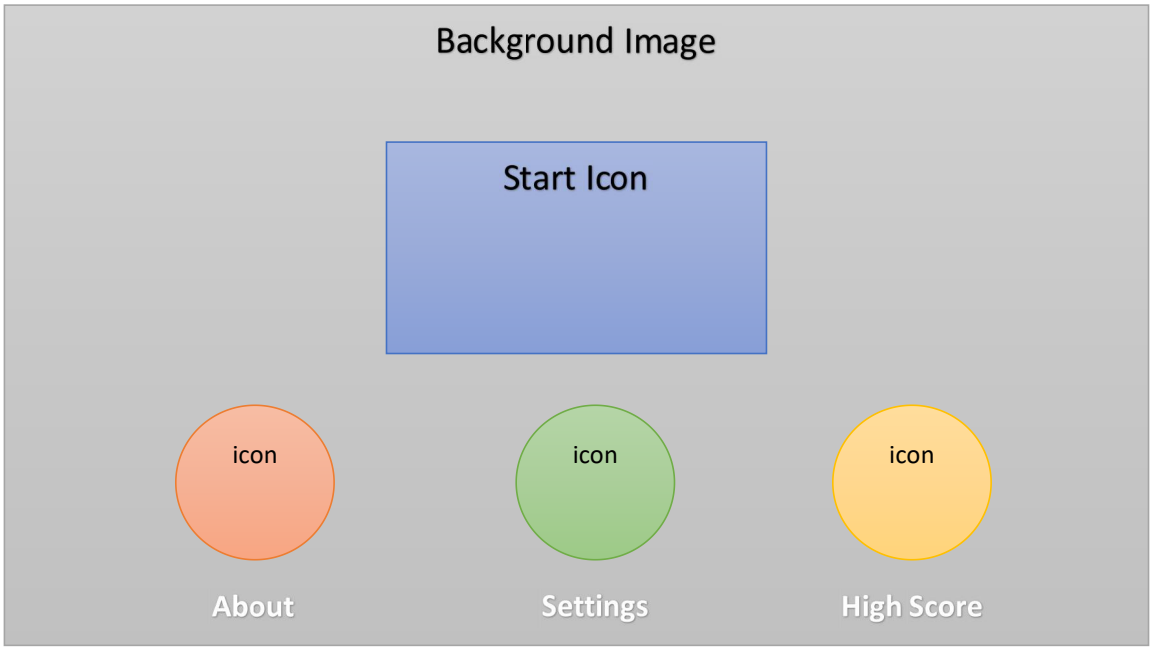


Figure 4.2: Start Screen story board

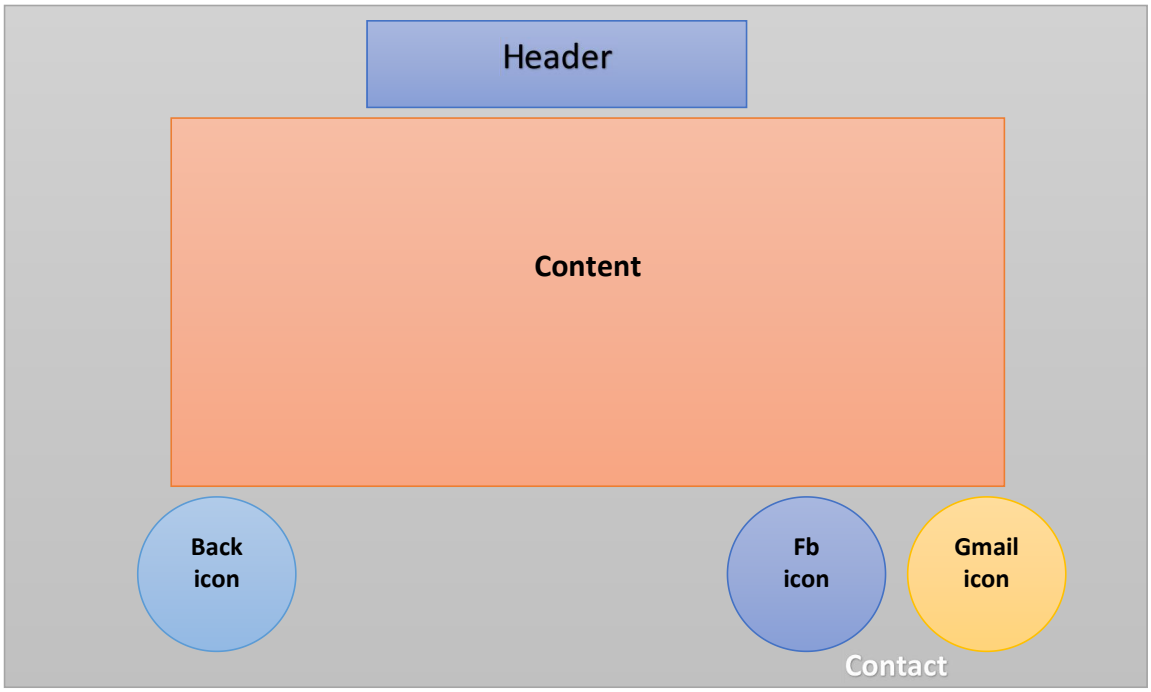


Figure 4.3: About Screen story board

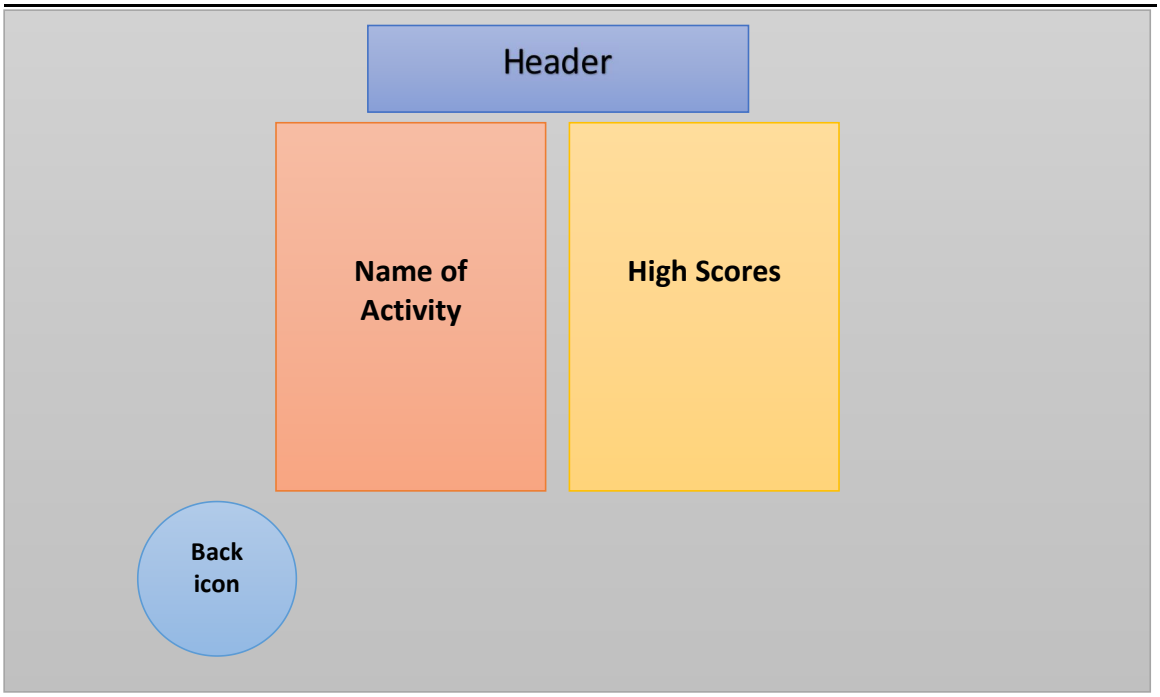


Figure 4.4: High score Screen story board

4.4. Navigational maps

The next step is of navigational maps. In these maps, the storyboards are used as an input. The different display buttons or action buttons show the navigation from one screen to the other. In other words when one action button is pressed it would lead to other screens. This path and navigation would be shown.

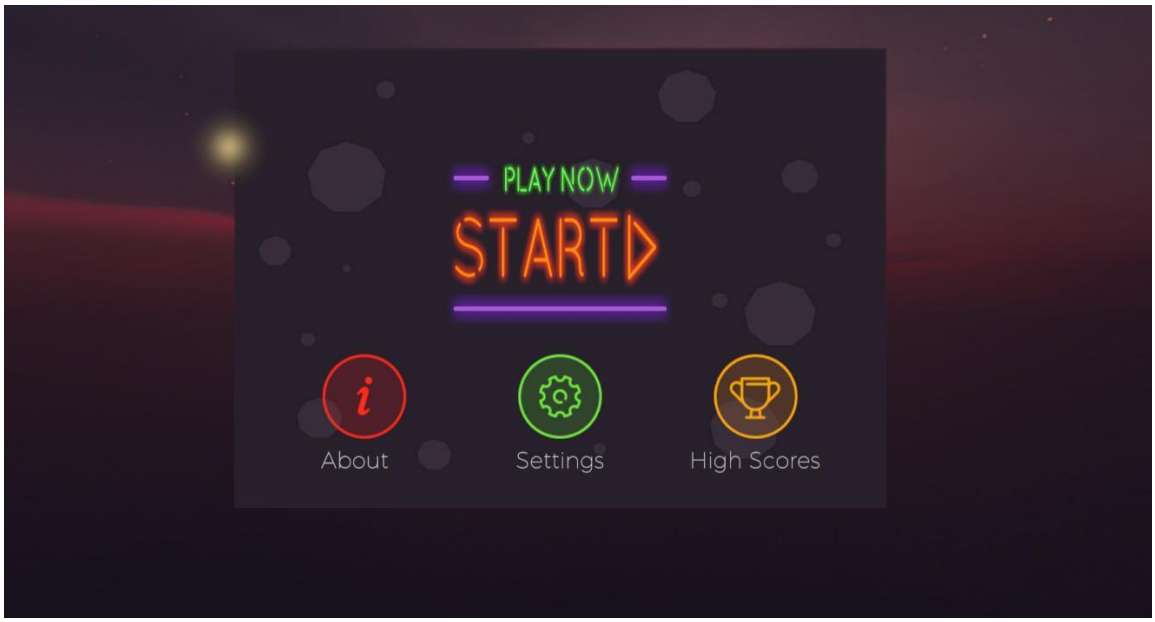


Figure 4.5: Start Screen UI

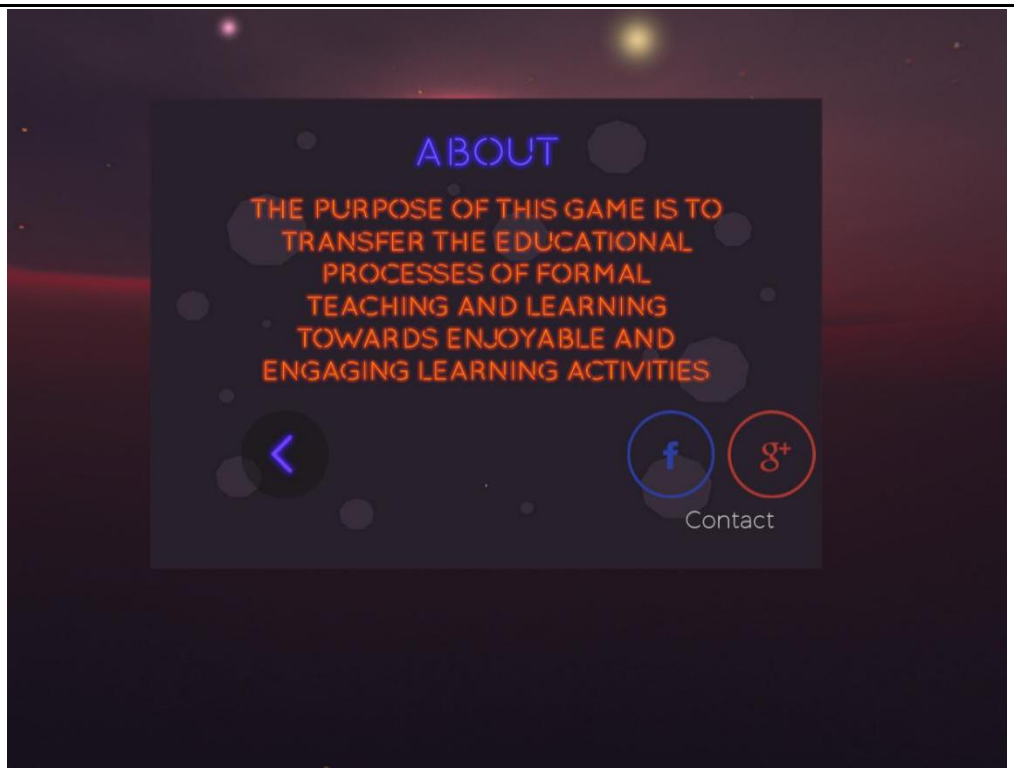


Figure 4.6: About Screen UI

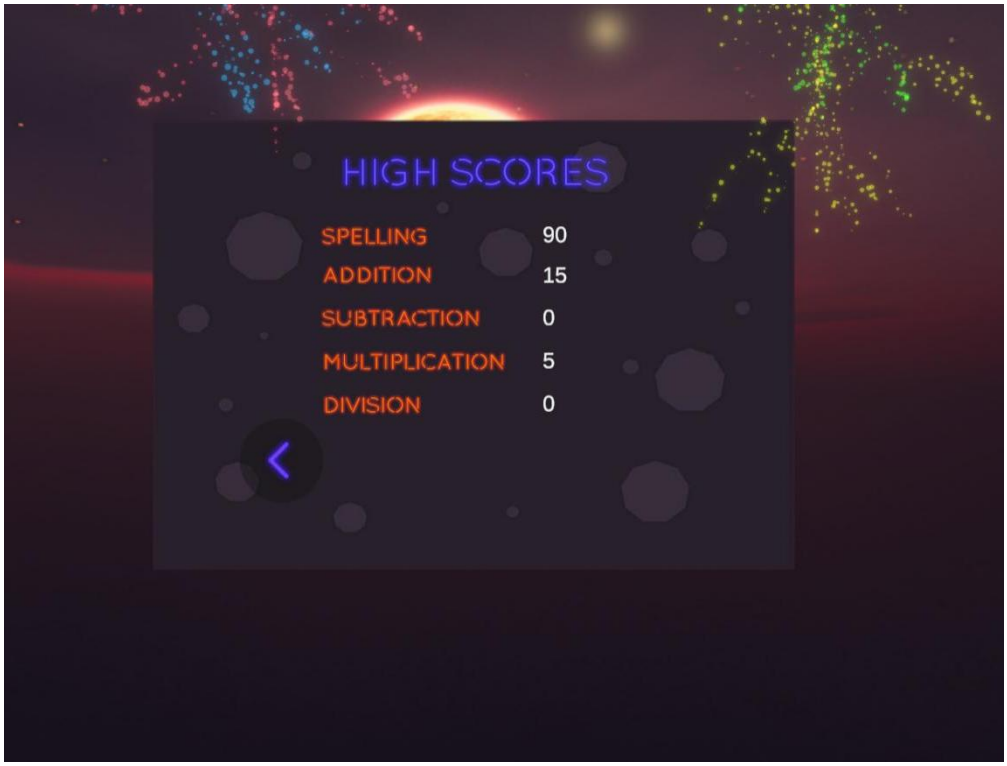


Figure 4.7: High Score UI

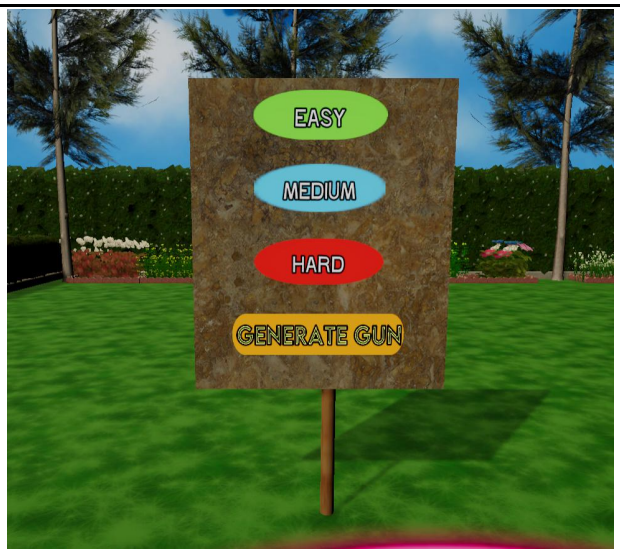


Figure 4.8: Spelling Activity UI

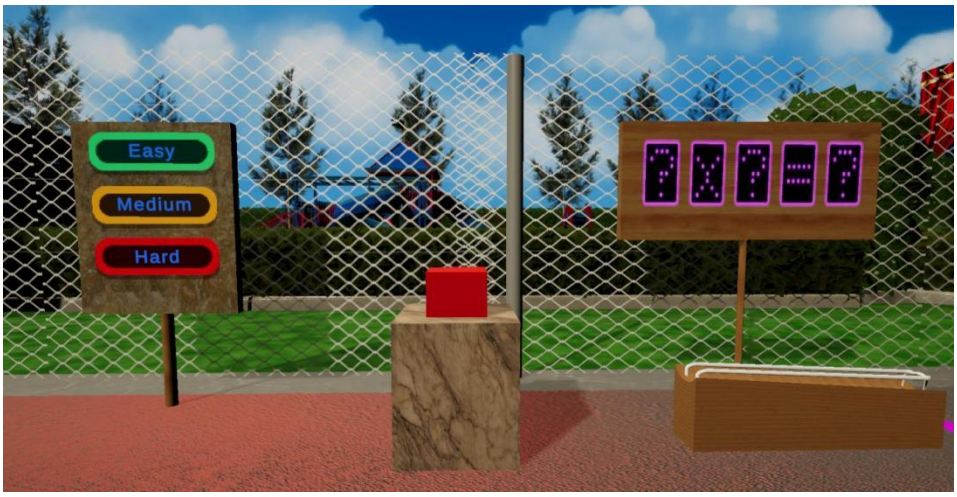


Figure 4.9: Basketball Activity UI



Figure 4.10: Color Activity UI

Chapter 5: Software Testing

5.1. Introduction

This deliverable is based on the IEEE standard of software testing i.e. IEEE SOFTWARE TEST DOCUMENTATION Std 829-1998. This standard describes a set of basic test documents that are associated with the dynamic aspects of software testing (i.e., the execution of procedures and code). The standard defines the purpose, outline, and content of each basic document. While the documents described in the standard focus on dynamic testing, several of them may be applicable to other testing activities (e.g., the test plan and test incident report may be used for design and code reviews). This standard may be applied to commercial, scientific, or military software that runs on any digital computer. Applicability is not restricted by the size, complexity, or criticality of the software. However, the standard does not specify any class of software to which it must be applied. The standard addresses the documentation of both initial development testing and the testing of subsequent software releases. For a particular software release, it may be applied to all phases of testing from module testing through user acceptance. However, since all of the basic test documents may not be useful in each test phase, the particular documents to be used in a phase are not specified. Each organization using the standard will need to specify the classes of software to which it applies and the specific documents required for a particular test phase.
The standard does not call for specific testing methodologies, approaches, techniques, facilities, or tools, and does not specify the documentation of their use. Additional test documentation may be required (e.g., code inspection checklists and reports). The standard also does not imply or impose specific methodologies for documentation control, configuration management, or quality assurance. Additional documentation (e.g., a quality assurance plan) may be needed depending on the particular methodologies used.
Following are standard artifacts, which must be included in this deliverable:

- Test Plan
- Test Design Specification
- Test Case Specification
- Test Procedure Specification
- Test Item Transmittal Report
- Test Log
- Test Incident Report
- Test Summary Report

5.2. Test plan

5.2.1. Purpose

The reason for this test is to detect the defect and errors in the game in early stage of development and fix it.
Scope: UI Interaction (buttons) , Scene Loading , Saving System , Locomotion , grab/throw
Tester: project Developer

5.2.2. Outline

A test plan shall have the following structure:

• Test plan identifier

• Introduction
• Test Items
• Features to be tested
• Features not to be tested
• Approach
• Item pass/fail criteria
• Suspension criteria and resumption requirements
• Test deliverables
• Testing tasks
• Environmental needs
• Responsibilities
• Staffing and training needs
• Schedule
• Risks and contingencies
• Approvals

The sections shall be ordered in the specified sequence. Additional sections may be included immediately prior to Approvals. If some or all of the content of a section is in another document, then a reference to that material may be listed in place of the corresponding content. The referenced material must be attached to the test plan or available to citizens of the plan. Details on the content of each section are contained in the following sub-clauses.

5.2.2.1. Test plan identifier

Unique identifier assigned to this test plan : **Basic_Features**

5.2.2.2. Introduction

In this testing phase the basic functionalities and high priority functionalities will be tested to check whether they are working or not. The basic functionalities are:
UI interaction: This is important because user will interact with it from the start of the game to start and interact with other components of the game. It include: Button working checking, Text fields updation etc.
Scene Loading: As the game will include different scenes , it will be very crucial that each scene gets loaded when needed. The loading of the scenes will be tested here.
Saving System: The saving system will be tested here to check whether the scores are being correctly stored in device memory (store) and then whether the load correctly or not.
Locomotion: One of the most crucial and important feature to test is locomotion. It this two locomotion techniques will be tested. First, **Teleportation** , this locomotion is very important as it is the mean to eliminate motion sickness. Here, it will be tested whether player is able to teleport from one place to another using the hand controllers or not. It will also be tested that that the place where player teleport should be accessible and not out of reach containing obstacles. Second, **Continuous** , this locomotion will also be tested using analog controls of the hand controller.
Grab/throw: Grab/ throw mechanism is also very crucial in which it will be tested that whether player is able to grab the interactable objects or not. And after grabbing, I=are they able to throw or not. This feature is very important for the sake of interactivity with game world environment.

5.2.2.3. Test items

Identify the test items including their version/revision level. Also specify characteristics of their transmittal media that impact hardware requirements or indicate the need for logical or physical transformations before testing can begin (e.g., programs must be transferred from tape to disk).
Supply references to the following test item documentation, if it exists:
a) Requirements specification
b) Design specification
c) Citizens guide
d) Operations guide
e) Installation guide
Reference any incident reports relating to the test items. Items that are to be specifically excluded from testing may be identified.

5.2.2.4. Features to be tested

Follow features will be tested:
UI Interaction (buttons) , Scene Loading , Saving System , Locomotion , grab/throw

5.2.2.5. Features not to be tested

All the features related to UI.

5.2.2.6. Approach

I will be using following approaches for this testing phase:
- Unit testing
- Integration testing
- System testing
- Acceptance testing

In Unit testing, I will be testing each feature individually. I will check whether these features are working alone or not. If these features pass this test, I'll move to Integration testing. In the approach I'll be seeing if the features still work perfectly after being integrated with other functionalities or not. I'll be looking the dependencies between these features and other functionalities. If all the integrated features according to the required standard, I'll be testing the entire system to see how the system works and feels. It will be tested that all the functionalities are working fine or not. If they work, I'll be testing them accordance to the standard and overall theme of the game. I'll see if they look good together or not. It will also be tested by keeping the player's point of view in accordance. Unit testing will take: 6,7 days. Integrating will take 3,4 days. System will take 2 days. And acceptance will take 1 day. In short, all should be completed within 2 weeks.

5.2.2.7. Item pass/fail criteria

After the testing, all the items are up to the required standard of game and have been declared pass.

5.2.2.8. Suspension criteria and resumption requirements

Some of the testing might require more time when other functionalities are added to the game. They may again require integrating and system testing again. They'll be resumed when all other functionalities and features have been added.

5.2.2.9. Test deliverable

Identify the deliverable documents. The following documents should be included:
a. Test plan;
a. Test design specifications;
b. Test case specifications;
c. Test procedure specifications;
d. Test item transmittal reports;
e. Test logs;
f. Test incident reports;
g. Test summary reports.
Test input data and test output data should be identified as deliverable.
Test tools (e.g., module drivers and stubs) may also be included.

5.2.2.10. Testing tasks

- They Game engine (Unity) should be working fine before testing.
- Oculus device should working fine with oculus link.
- There should be some other players, for acceptance test.
- Record the experience while playing so that it can be seen later.

5.2.2.11. Environmental needs

As, the project is related to VR, the environment should be vacant without any furniture so that the tester couldn't get hurt while testing procedure. There can also be a screen where the experience of the tester could be seen by any other person. Other than that, there are no such environmental needs.

5.2.2.12. Responsibilities

Identify the groups responsible for managing, designing, preparing, executing, witnessing, checking, and resolving. In addition, identify the groups responsible for providing the test items identified and the environmental needs identified.
These groups may include the developers, testers, operations staff, citizen representatives, technical support staff, data administration staff, and quality support staff.

Table 5.1: Responsibilities table

Testing Module	Performed By
Testing Planning	Faryal Azhar
Test Specification	Faryal Azhar
Test Case Development	Faryal Azhar
Test Writing	Faryal Azhar
Test Execution	Faryal Azhar

5.2.2.13. Staffing and training needs

The tester should know about Oculus VR headset and its controllers. The tester should know, which button is used for grabbing, teleporting etc.

5.2.2.14. Schedule

Unit testing will take: 6,7 days. Integrating will take 3,4 days. System will take 2 days. And acceptance will take 1 day. In short, all should be completed within 2 weeks.
This testing phase will be completed within 2 weeks. When more functionalities are added into game, this phase testing will resume again for day or two.

5.2.2.15. Risks and contingencies

The main risk in the regularity of schedule is the unavailability of needed resources for testing. I.e VR headset. Other risks are , delay in functionality implementation. In this case, test items might require increased night shift scheduling to meet the delivery date.

5.2.2.16. Approvals

Table 5.2: Approval for Staffing and training needs

Signature	
Name	Faryal Azhar
Role	Project Developer
Date	25 June, 2021

5.3. Test design specification

5.3.1. Purpose

The reason for this test is to see whether the theme and other User-interface elements are upto the standards or not. They'll be tested in accordance to the taste of player.
Scope: Main menu screen, About screen, High Score screen, Color board
Tester: project Developer

5.3.2. Outline

A test plan shall have the following structure:
a. Test plan identifier;
b. Introduction;
c. Test items;
d. Features to be tested;
e. Features not to be tested;
f. Approach;
g. Item pass/fail criteria;
h. Suspension criteria and resumption requirements;
i. Test deliverable;
j. Testing tasks;
k. Environmental needs;
l. Responsibilities;
m. Staffing and training needs;
n. Schedule;
o. Risks and contingencies;
p. Approvals.

The sections shall be ordered in the specified sequence. Additional sections may be included immediately prior to Approvals. If some or all of the content of a section is in another document, then a reference to that material may be listed in place of the corresponding content. The referenced material must be attached to the test plan or available to citizens of the plan. Details on the content of each section are contained in the following sub-clauses.

5.3.2.1. Test plan identifier

The unique identifier assigned to this test plan: UI_Design

5.3.2.2. Test items

Front end designing will be tested in this phase of testing. I will check whether the theme and coloring scheme is according to the requirement of game or not.

a. Requirements specification
b. Design specification
c. Users guide
d. Operations guide
e. Installation guide

Reference any incident reports relating to the test items. Items that are to be specifically excluded from testing may be identified.

5.3.2.3. Features to be tested

All the features related to User Interface will be tested. They will be tested accordance to the theme of game. The UI will be tested from the point view of a player (in this case kid). It will be noted that whether the kid like the interface and finds it interesting/exciting or

not. The technical detail will not be tested. If any issue find here, it will be rated low,medium or high according to their priority.
For my project, features to be tested are:

- Main menu screen
- About screen
- High Score screen
- Color board
- Difficulties board

5.3.2.4. Features not to be tested

All the technical features (I.e whether the buttons are working or not etc)

5.3.2.5. Item pass/fail criteria

All the items have been passed, they are accordance to the taste of player. None is failed.

5.3.2.5. Approvals

Table 5.3: Approval for Test design

Signature	
Name	Faryal Azhar
Role	Project Developer
Date	25 June, 2021

5.4. Test case specification

Table 5.4: Test case (haptic feedback)

Test Engineer	Faryal Azhar
Test case ID	TC1
Purpose	Player should feel feedback on hands in the form of vibration while interacting with things.
Pre-Requisition	1. The hand controllers and their sensors are working properly. 2. The player should interact with any object.
Steps	1. Start the game. 2. Try to interact with objects by grabbing the objects.
Test Identification	TI1
Status	Pass

Table 5.5: Test case (save)

Test Engineer	Faryal Azhar
Test case ID	TC2
Purpose	The score of activity gets save and stores in device memory whenever player makes high-score.
Pre-Requisition	The player should be playing activity and able to make high-score.
Steps	1. Starts the game. 2. Play any activity of choice. 3. Scores highest.
Test Identification	TI2
Status	Pass

Table 5.6: Test case (sound effects)

Test Engineer	Faryal Azhar
Test case ID	TC3
Purpose	The sound effects (sfx) should play to give the appropriate feedback to user in form of sound (for example, ball collision sound).
Pre-Requisition	1. The VR headset's speaker should be working fine. 2. To listen the sfx, player should perform or play any activity. (throw ball).
Steps	1. Start the game 2. Play any activity. 3. Interact with Objects.
Test Identification	TI3
Status	Pass

Table 5.7: Test case (visual effects)

Test Engineer	Faryal Azhar
Test case ID	TC4
Purpose	The visual effects (vfx) should play to give the appropriate feedback to user in form of visuals (for example, the fireworks on correct guess).
Pre-Requisition	To see the vfx, player should perform or play any activity. (make a correct guess, throw ball etc).
Steps	1. Start the game. 2. Play any activity. 3. Interact with Objects. 4. Provide answers in activity.
Test Identification	TI4
Status	Pass

Table 5.8: Test case (load)

Test Engineer	Faryal Azhar
Test case ID	TC5
Purpose	When player starts the game, the high-score gets automatically loaded into game.
Pre-Requisition	There should be already saved high-score file.
Steps	1. Start the game and go to main menu's "High-score" option. 2. There, should be the high-score opposite to each activity. 3. Play the game activity 4. The previously scored high-score gets Loaded and over-written if player score higher.
Test Identification	TI5
Status	Pass

Table 5.9: Test case (teleportation)

Test Engineer	Faryal Azhar
Test case ID	TC6
Purpose	The player should be able to teleport from one place to another using VR controller.
Pre-Requisition	1. The game should be started and controllers should be working properly. 2. The targeted place to teleport should be accessible and should have no obstacles.
Steps	1. Start the game. 2. Then aims the teleportation line on the accessible area. 3. Press the button on controller.
Test Identification	TI6
Status	Pass

Table 5.10: Test case (grab)

Test Engineer	Faryal Azhar
Test case ID	TC7
Purpose	The player should be able to grab the objects in game. The grabbing action will be performed with hand controllers, using grab button. The sensitivity of grab button will also affect the mechanism.
Pre-Requisition	1. The controllers should be working properly. 2. The player should know the grab button.
Steps	1. Start the game 2. Take hand towards the target object 3. Press grab button.
Test Identification	TI7
Status	Pass

Table 5.11: Test case (difficulty level)

Test Engineer	Faryal Azhar
Test case ID	TC8
Purpose	The player should be given choice to select the difficulty level of the activities. There should be three levels : Easy, Medium and Hard. The easy level should increment score with small margin, medium should increase with medium margin and hard mode should increase the scores with highest margin.
Pre-Requisition	The player is in activity area. The player is interacting with difficulty board by standing in front of it.
Steps	1. Go to any activity area in garden. 2. Stand in front of interactive difficulty board. 3. Move the hand to the desired difficulty level and press the select button. 4. The activity gets easier or harder according to the choose option.
Test Identification	TI8
Status	Pass

Table 5.12: Test case (scene loading)

Test Engineer	Faryal Azhar
Test case ID	TC9
Purpose	The game scene should be loaded when select start game in main menu.
Pre-Requisition	Should be on start screen (on main menu)
Steps	1. Start the game. 2. Take interaction line on “Start Game” 3. Press select button.
Test Identification	TI9
Status	Pass

5.5. Test procedure specification

5.5.1. Purpose

To specify the steps for executing a set of test cases or, more generally, the steps used to analyze a software item in order to evaluate a set of features.

5.5.2. Outline

A test procedure specification shall have the following structure:

a. Test procedure specification identifier
b. Purpose

c. Special requirements
d. Procedure steps

5.5.2.1. Test procedure specification identifier

The identifier set for procedure is “EVRG”.

5.5.2.2. Purpose

To define a test procedure specified by a test design specification and test case specification.

5.5.2.2. Special requirements

As discussed in test case specification’s prerequisite.

5.6. Test Summary report

5.6.1. Purpose

To sum up the outcome of the selected testing actions and to afford evaluation based on these results.

5.6.2. Outline

A test summary report shall have the following structure:

a. Test summary report identifier
b. Summary
c. Comprehensive assessment
d. Summary of results
e. Evaluation
f. Summary of activities
g. Approvals

The sections shall be ordered in the specified sequence. Additional sections may be included just prior to Approvals. If a few or all of the contented of a part is in one more documents, then a position to that substance may be listed in place of the matching contented. The referenced substance must be fond of to the test summing up report or accessible to users of the summary report.
facts on the contented of each segment are limited in the following sub clauses.

5.6.2.1. Test summary report Identifier

Specify the unique identifier assigned to this test summary report.

5.6.2.2. Summary

All the test cases passed their respective tests. They were updated and tested again to meet the specified requirement. In short,all the test cases were tested against their requirement and all the requirements were met. The code was then optimized to get better

result. The test was performed using Oculus quest 1, in an empty space (with no furniture) and good lights. For testing accurately the logs were used.

5.6.2.3.Variances

No, there were no variances of test items from their design specification. There were no variances from their test plan, test designs, or test procedures either. All the tests were performed according to the plan with specified time span.

5.6.2.4. Comprehensiveness assessment

Evaluate the comprehensiveness of the testing process against the comprehensiveness criteria specified in the test plan if the plan exists. Identify features or feature combinations that were not sufficiently tested and explain the reasons.

5.6.2.5. Summary of results

All the test cases passed their respective tests. They were updated and tested again to meet the specified requirements. There were no unresolved incidents.

5.6.2.6. Evaluation

Provide an overall evaluation of each test item including its limitations. This evaluation shall be based upon the test results and the item level pass/fail criteria. An estimate of failure risk may be included.

5.6.2.7. Summary of activities

Summarize the major testing activities and events. Summarize resource consumption data, e.g., total staffing level, total machine time, and total elapsed time used for each of the major testing activities.

Table 5.13: Summary of activities

Functions	Description	%TC executed	%TC passed	%TC pending	Priority	Remarks
Haptic feedback	Player should feel feedback on hands in the form of vibration while interacting with things.	100	100	0	High	passed
Save	The score of activity gets save and stores in device memory whenever player makes high-score.	100	100	0	High	passed
Sound_Effects	The sound effects (sfx) should play to give the appropriate feedback to user in form of sound (for example, ball collision sound).	100	100	0	Normal	passed
Load	When player starts the game, the high-score gets automatically loaded into game.	100	100	0	High	passed
Visual_effects	The visual effects (vfx) should play to give the appropriate feedback to user in form of visuals (for example, the fireworks on correct guess).	100	100	0	Normal	passed
Teleport	The player should be able to teleport from one place to another using VR controller.	100	100	0	High	passed
Grab	The player should be able to grab the objects in game. The grabbing action will be performed with hand controllers, using grab button. The sensitivity of grab button will also affect the mechanism.	100	100	0	High	passed

5.6.2.8. Approvals

Table 5.14: Approvals

Signature	
Name	Faryal Azhar
Role	Project Developer
Date	28 June, 2021

Chapter 6: User Manual

6.1.General Information

This game is developed to be used as a tool for providing the basic education to kids and making the learning experience enjoyable with the factor of immersion and interaction. While playing the game, the user will be asked to solve different educational activities in a fun way. The game will also have various difficulties level so that the activities can be made harder as per user requirement. The project is developed by combining the game and education which will encourage the creative thinking in kids.
Game can be played on Oculus quest with hand-controllers (might work on all Oculus VR headset, haven't tested). VR technology for this project, it will provide us with the benefit of immersion and interaction. As the studies have shown, the children can learn better with interaction. So, the project will prove very beneficial for everyone (kids, teachers, parents etc).

6.1.1. System Overview

The main purpose "Educational VR game for kids" is to transfer the educational process of formal teaching and learning towards enjoyable and engaging learning activities. While playing this game, kids can learn colors, spellings, mathematical operations (addition, subtraction, multiplication, division). It can also be expanded in future to have even more activities.

6.2.System Summary

The developed game will work on Oculus quest 1 (VR headset). It can also be programmed to work on other Oculus VR headset with a little bit of code and change.

6.2.1. System Configuration

The game will be published on Oculus app store in future after working and testing thoroughly on project with code optimization. The payer will then download it from there. Game doesn't need internet access to work, it will work offline.

6.2.2. User Access Levels

Game will have only one access level for user, which is "player". Player will be allowed to play game anytime after installing it from store. Other than that, there will be no special access levels or extra permissions given to the users.

6.2.3. Contingencies

In the case of any exception or error, player will be allowed to send an email to the developer.

6.3.Getting Started

Download the game from oculus app store and start the game.

6.3.1. Usage Guide

To play the game, user must have Oculus Quest working properly.

6.3.2. Game Controls

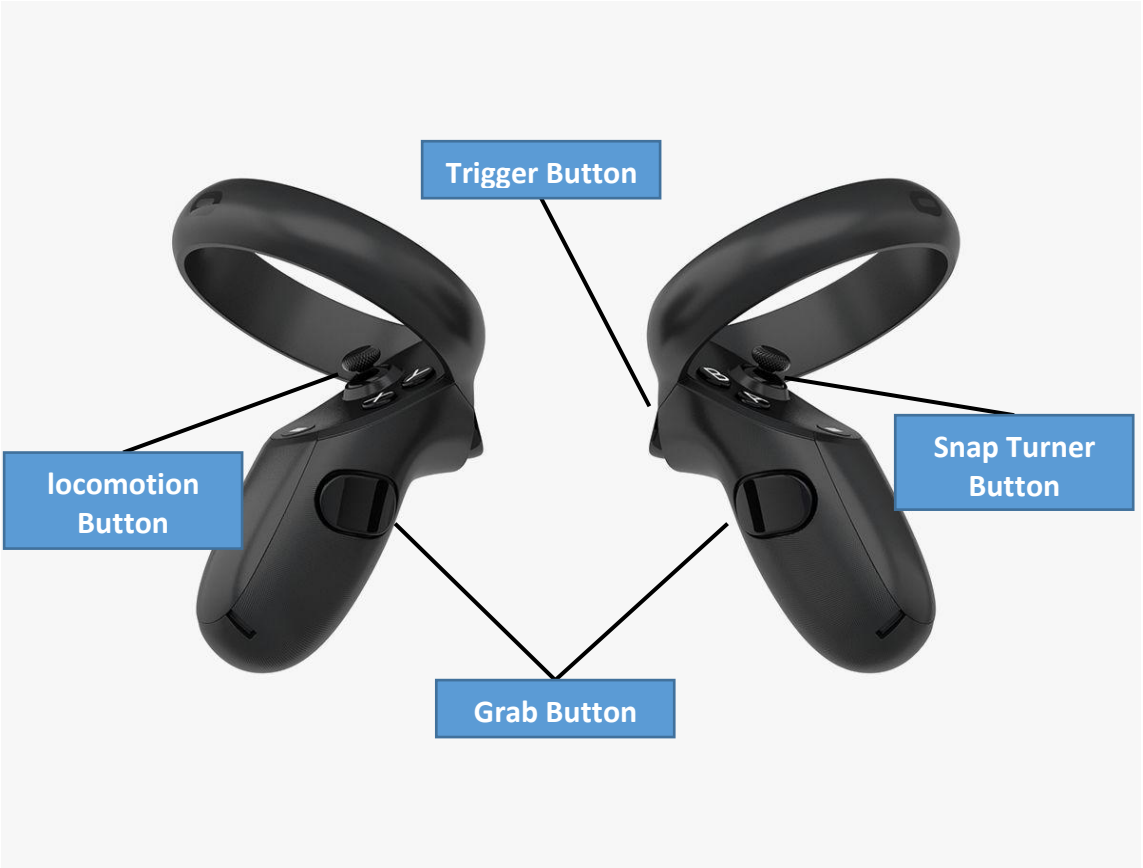


Figure 6.1: Oculus hand-controllers controls



Figure 6.2: Oculus headset controls

6.3.2.1. Button Usage

- **Trigger Button :**
  a. To teleport from one place to another.
  b. To interact with UI (select options).
  c. To fire.

- **Snap turner Button :**
  a. To snap rotate 45 degree

- **Locomotion Button :**
  a. Analog button for Continuous movement (locomotion)

- **Grab Button :**
  a. To grab objects.
  b. To open door and drawers etc.

6.4. Using System

The game consist of two scenes. Scene one is “Start” scene, and other is “Game” Scene. The scene which loads at the beginning of game is start scene.

6.4.1. Start Scene

Start screen will provide the user to access some of the main options/screens of game. It is also vital because, through start scene, player will start/play the game. Start Scene provides: About (which will have info regarding the game and developer. User can see the about screen to contact the developer through email or social media), Setting (which will include music and sound effects settings), High Score (which will hold the high scores scored by player). Player will interact with these screens by “trigger” button.

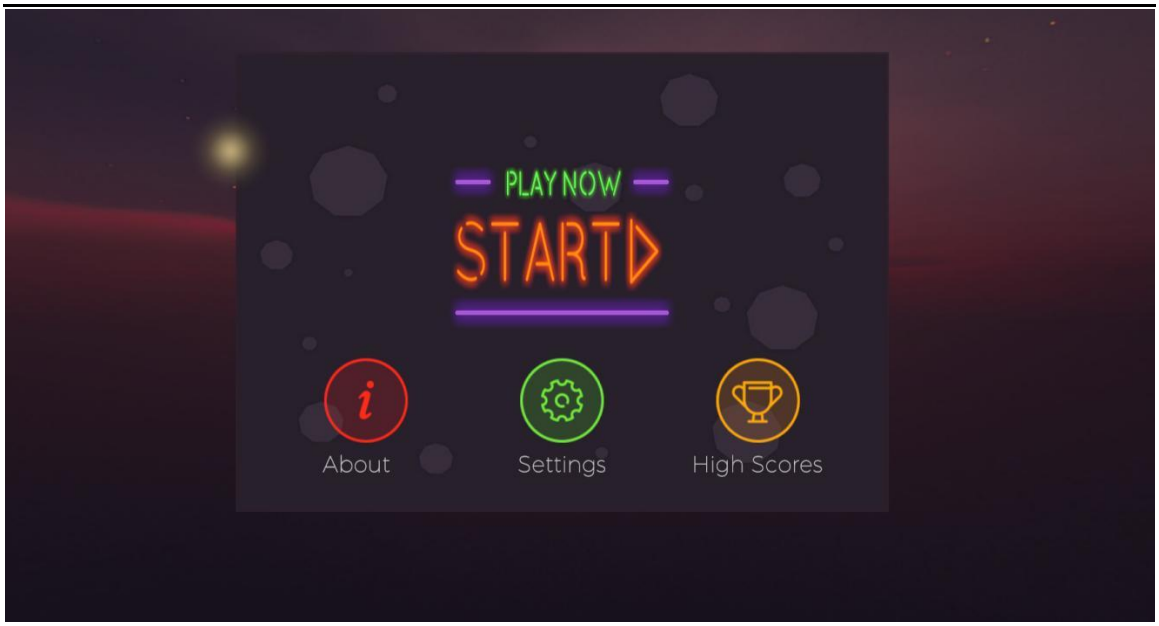


Figure 6.3: Start Screen

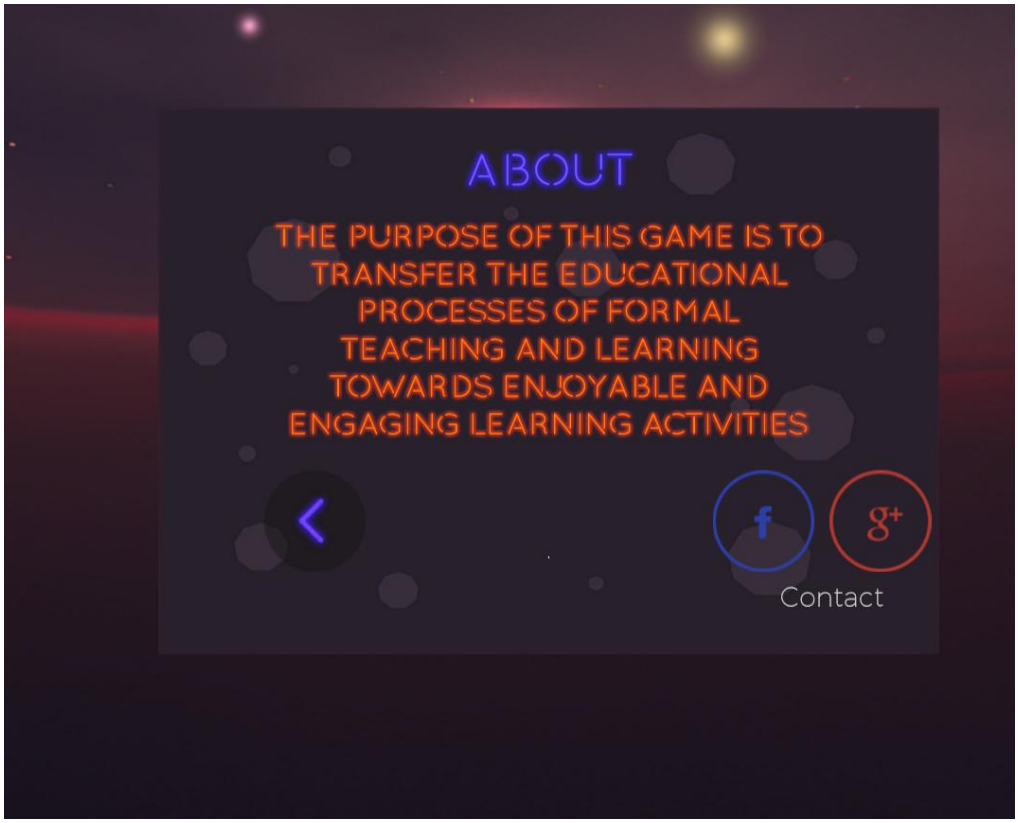


Figure 6.4: About Screen

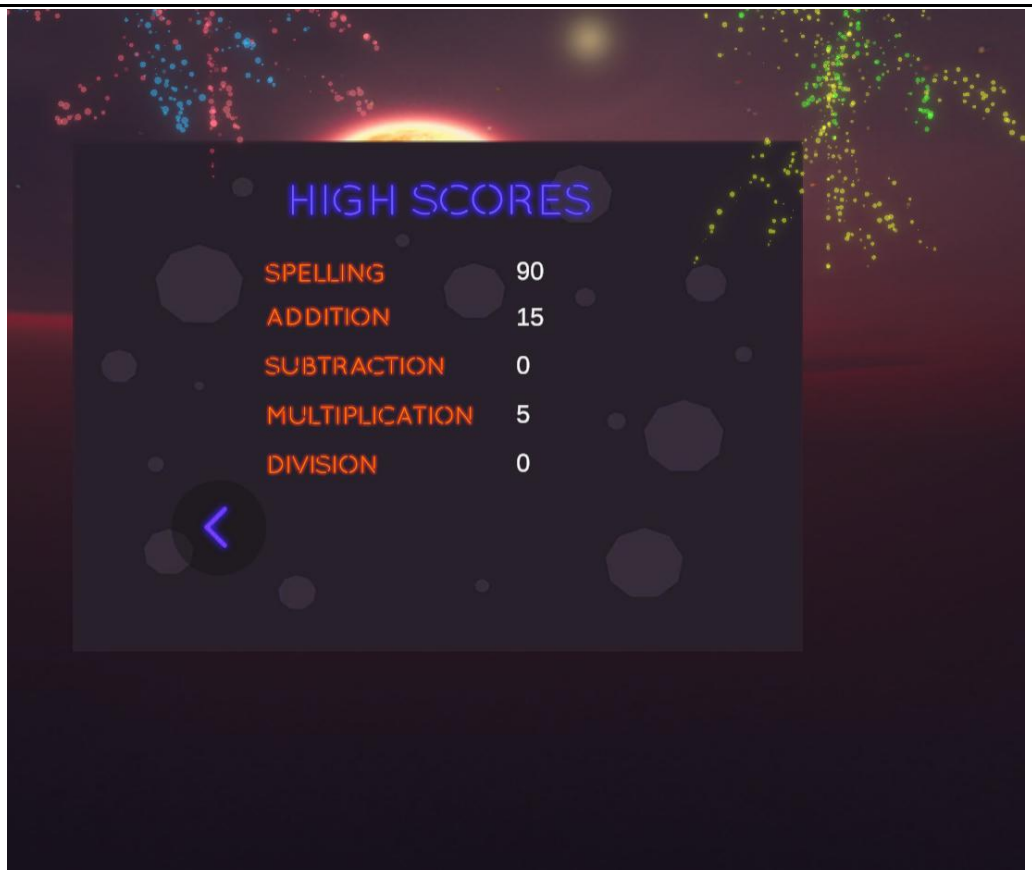


Figure 6.5: High Score Screen

6.4.2. Room and Color Activity

When player selects “Start”, the game will start and player will be inside the room with door locked.



Figure 6.6: Locked door

Player will be asked to “clean” the room by completing a “color activity”. Player will be asked to choose three colors of their choice. After choosing the colors, three buckets of those color will appear in front player, and nine blocks will also appear (grouped into three colors, each color will contain 3 blocks).



Figure 6.7: Color Selection

Player will then have to put all the blocks into their respective buckets to complete the activity and unlock the door. Once, activity will be completed, player will be allowed to go in the garden.

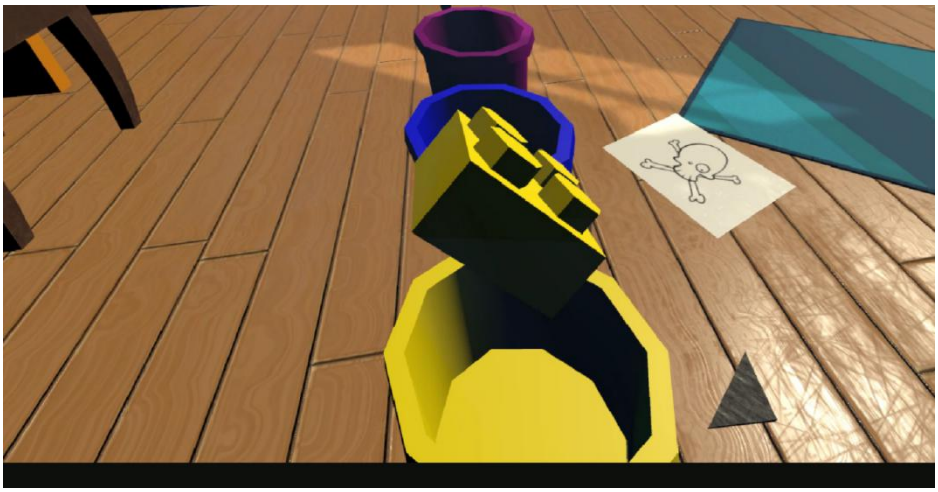


Figure 6.8: Blocks and buckets



Figure 6.9: Garden Entrance

6.4.3. Basketball Activity (Mathematical Operations)

On the right bottom side of the garden, there will be "Basketball Activity". It consist of for section.

- Green Section (For Addition)
- Orange Section (For Subtraction)
- Red Section (For Multiplication)
- Blue Section (For Division)

Player can go in any section and practice their mathematical skills.



Figure 6.10: Addition and Multiplication site overview



Figure 6.11: Subtraction and Division site overview

After going to section, the player will need to press the button to start the activity. On start, button will have "Red" color (which means that button is off), when player will press button, button will turn "Green" (which means the activity has been started).

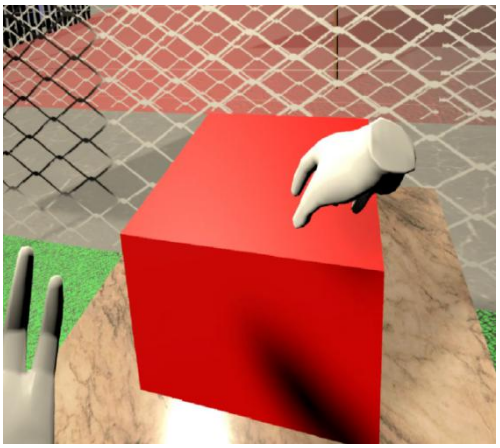


Figure 6.12: Basketball button off

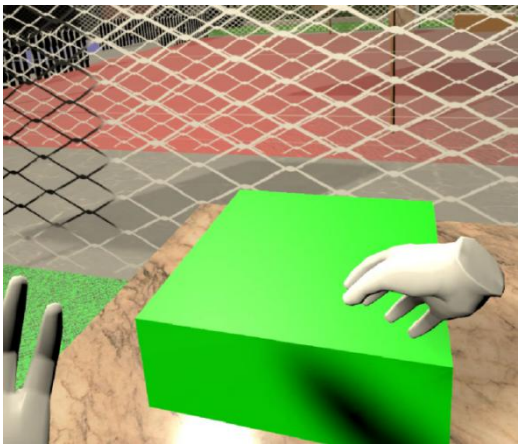


Figure 6.13: Basketball button on

After pressing the button, the ball will be generated on ball-stand and question will appear on question board.



Figure 6.14: Basketball question board

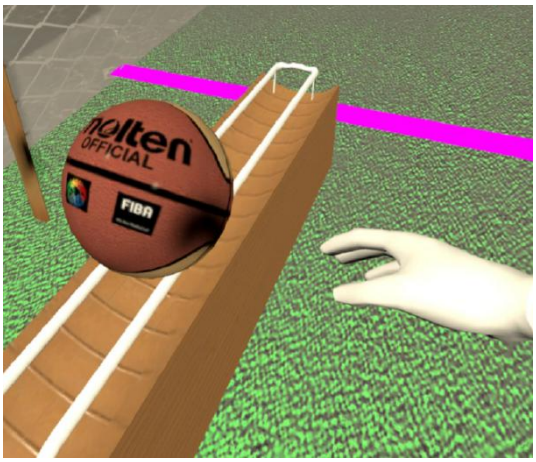


Figure 6.15: Basketball activity's ball

After reading the question, player will have to grab ball up and score it into its correct basket by looking at the value assigned to each basket. The score of player will go up, if they score the ball into correct basket.



Figure 6.16: Basket's random number assignment

6.4.4. Spelling Activity

The Spelling activity is located on top-left part of garden. When player reach there, to start the activity, player will need to select start button located on the bottom of stage.

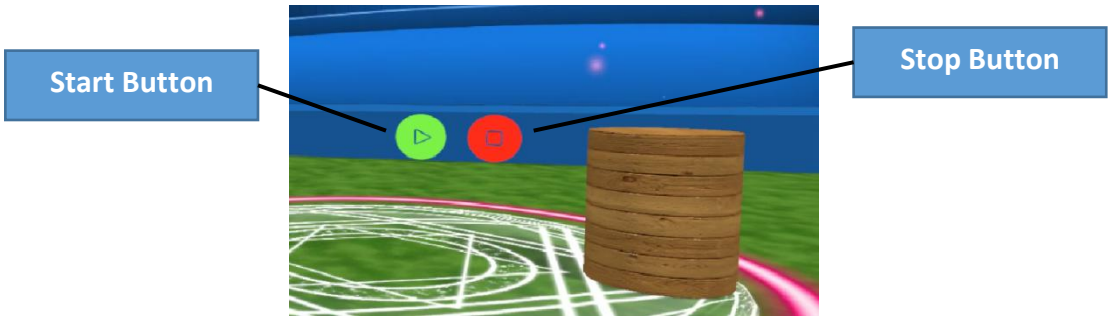


Figure 6.17: Spelling activity start and stop button

After starting the activity, the scrambled word will appear on the screen of the stage. The balloons with one character each, will also appear in front of player.

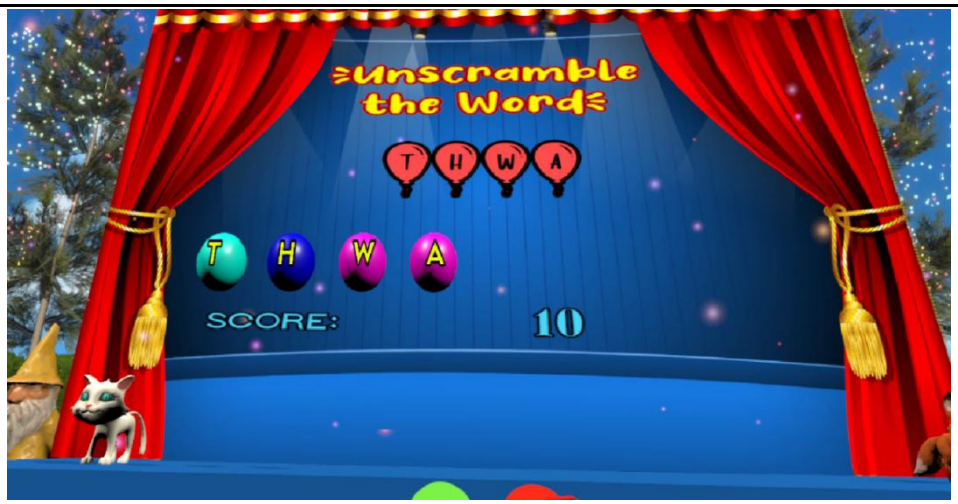


Figure 6.18: Spelling Activity Scrambled word

Then, player will have to unscramble the word, by shooting balloons in a correct order with the help of laser gun. Player can generate laser gun by selecting "Generate Gun". To pickup the gun, player will have to bring their hand closer to the gun and press "Grab" button. To fire the laser, player will need to press "trigger" button.



Figure 6.19: Spelling Activity's Laser gun

If player is able to unscramble the word correctly, the score of player will be increased.



Figure 6.20: Spelling activity unscrambled word and score increase

6.4.5. Select Difficulty

Player will be given three difficulty levels for each activity i.e Easy, Medium, and Hard. Player can choose their required level, by selecting the option on the difficulty board (by pressing “trigger” button).



Figure 6.21: Basketball activity difficulty selector



Figure 6.22: Spelling activity difficulty selector

Appendixes

Appendix 1:

Abbreviations:

VR: Virtual Reality

Unity:
Unity is a cross-platform game engine and world's most popular development platform for creating 2D and 3D multiplatform games and interactive experiences.

Oculus Quest:
Oculus Quest is a virtual reality headset developed by Oculus, a division of Facebook, released on May 21, 2019. It is a standalone device that can run games and software wirelessly under an Android-based operating system.

Visual Studio:
Microsoft Visual Studio is an integrated development environment from Microsoft. It is used to develop computer programs, as well as websites, web apps, web services and mobile apps.

Sourcetree:
Sourcetree is a Git GUI that offers a visual representation of your repositories. It is a free Git client for Windows and Mac.

Oculus (Software):
Oculus is used to manage the oculus vr devices and explore apps in oculus store, discover live VR events and so much more. During this project, Oculus is used on windows to link the VR headset to the PC.

Git:
Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.